Continuous Hidden Markov Models for Equity Returns: Heavy-Tail Emission Families and Regime-Conditional Value-at-Risk

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Abstract

Synthetic generators of daily US-equity returns support stress testing, regulatory backtesting, and downstream model training on data subject to privacy or licensing constraints, but the regime-switching route has been shaped for two decades by a negative result: low-state-count hidden Markov models with Gaussian emissions fail to reproduce the slow ACF decay of absolute returns, the third of the Cont stylized facts. We revisited that result through a continuous hidden Markov model (CHMM) in which a textbook mixture-of-eigenvalues identity recasts the failure as a rank condition on the centred transition matrix; an empirical diagnostic showed the implied bound is not active at the cross-ticker median once enough latent states are admitted, shifting the limit from the chain's temporal structure to the marginal mixture. We developed a unified expectation-conditional-maximisation framework supporting four heavy-tailed emission families (Gaussian, Student-t, Laplace, and Generalised Error Distribution) under shared forward-backward recursions and quantile-based initialisation, with the maximisation step the only difference between variants. The empirical study covered SPY, a sector-balanced 30-ticker panel, a CRSP cross-decade transfer, and a six-asset copula extension. At $K^\star = 3$, the four emission families reproduced all three symmetric Cont stylized facts on the SPY in-sample window, with the shared-ν Student-t at simulated in-sample kurtosis 4.68 against observed 7.68 without a tuning hyperparameter. A regime-conditional Value-at-Risk built by propagating the one-step-ahead state forecast through the family-appropriate predictive mixture passed the Christoffersen joint conditional-coverage test on the main window and on 19/24 walk-forward rows, with an Engle-Manganelli dynamic-quantile backstop as the strict-tail test. The Student-t copula on a six-asset US-equity basket gave an in-sample off-diagonal correlation MAE of 0.027, against 0.076 for the single-index alternative. The construction is positioned against an i.i.d. bootstrap and a maximum-likelihood hidden semi-Markov benchmark that exceed it on raw single-window distributional fit; the CHMM is useful for things those alternatives cannot do. The scope is daily US equities in stable out-of-sample periods, and we recommend periodic refit for stress folds the static fit cannot bracket.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Value-at-Risk, Synthetic-Data Generator.

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1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of machine-learning models on data that cannot be released for privacy or licensing reasons [1–3]. A useful generator has to reproduce, at the same time, the standard stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [4, 5]. Generalised autoregressive conditional heteroskedasticity (GARCH) models [6, 7] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but lose temporal persistence; deep generators [8–10] reproduce some stylized facts at the cost of interpretability and reproducibility. One negative result has shaped two decades of work on regime-switching models. Rydén et al. [11] fitted Gaussian-emission hidden Markov models (HMMs) at low state counts and concluded that the geometric sojourn-time distribution of a standard Markov chain makes the ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] recovered the ACF fit by replacing the Markov chain with a hidden semi-Markov chain, at the cost of much greater complexity. The Rydén et al. failure combines two constraints: a temporal one (the absolute-return ACF must allow slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed kurtosis). A standard closed-form identity for the absolute-return ACF, written as a sum over the non-unit eigenvalues of the transition matrix [13–15], bounds the number of decay modes by the rank of the centred transition matrix. We ask whether that bound actually matters on equity-return data once enough latent states are allowed.

In this study, we built three things on top of the continuous HMM (CHMM): a unified expectation-conditional-maximisation (ECM) framework for the four emission families, in which only the M-step changes between variants; a regime-conditional Value-at-Risk (VaR) that averages the predictive density over the one-step-ahead state forecast; and a Student-t copula that ties the per-asset CHMM marginals into a joint generator through Sklar's theorem. We tested the model four ways: a six-fold rolling-origin walk-forward on the SPY decade, a sector-balanced 30-ticker panel across ten Global Industry Classification Standard (GICS) sectors, a Center for Research in Security Prices (CRSP) cross-decade transfer, and a six-asset US-equity basket, which together guard against single-window over-fitting. The four-family panel reproduced all three symmetric Cont stylized facts at the chosen $K^{\star}$, and heavy-tailed emissions narrowed the kurtosis gap relative to the Gaussian baseline. The rank bound from the bilinear ACF identity did not bind at the cross-ticker median, though it still mattered for a few tickers that introduced new regimes. The regime-conditional VaR passed the Christoffersen joint conditional-coverage test cleanly on the main window and on most walk-forward folds, and the Student-t copula matched off-diagonal correlations far better than the single-index alternative. We read this to mean that the Rydén low-K failure is a limit on the marginal mixture, not on the chain's temporal structure, once initialisation is fixed and the emissions can carry heavy tails. The i.i.d. bootstrap and a maximum-likelihood hidden semi-Markov model (HSMM) beat the CHMM on raw single-window distributional fit, but neither supports the regime-conditional risk forecasting, multi-asset copula composition, or parametric synthetic-data privacy the CHMM is built for. The scope is daily US equities: a non-equity stress test on the GLD exchange-traded fund (ETF) breaks down under static fitting, and we recommend periodic refit within that scope.

2 Related Work

The discrete-state regime-switching tradition opens with Hamilton [16], who introduced Markov-switching to economics, and Schaller and Van Norden [17], who estimated regime-switching specifications on US monthly stock returns and documented the predictive content of the latent regime for return forecasting. The main evaluation against the Cont stylized facts comes from Rydén et al. [11], who fitted Gaussian-emission hidden Markov models at $K = 2$ or 3 states and showed that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay of absolute returns. Bulla and Bulla [12] recovered the fit with hidden semi-Markov models at much higher complexity, replacing the geometric sojourn with an explicit-duration family at the cost of a duration-augmented forward-backward, and Nystrup et al. [18] extends the same line to long-memory volatility. Each of these works makes the same kind of structural choice we make here: the latent regime is a finite-state object with per-state emission densities, and what varies across the lineage is the sojourn distribution and the per-state family. The emission families that supply the per-state densities used in this work draw on the maximum-likelihood Student-t formulations of Peel and McLachlan [19] and Liu and Rubin [20], with the Generalised Error Distribution (GED) [21, 22] providing a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degree-of-freedom parameter ν_k plays in CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [3], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism, and the present continuous-emission CHMM replaces the discretisation step with the four per-state emission families above under a unified ECM framework, recovering the slow-ACF target through the eigenvalue argument rather than through an explicit duration model.

Conditional-variance and stochastic-volatility models form a parallel line in which the latent state is the volatility process rather than a discrete regime. The GARCH framework [6, 7] encodes volatility persistence through a single decay parameter and a unimodal innovation distribution and fails the Kolmogorov-Smirnov test on equity returns, and successive extensions have closed some gaps without removing this limit. Heavy-tailed innovations [23], asymmetric leverage variants (exponential GARCH, EGARCH, 24; Glosten-Jagannathan-Runkle GARCH, GJR-GARCH, 25; threshold GARCH, 26), regime-switching extensions (Markov-switching GARCH, MS-GARCH, 27, with the `MSGARCH` R package of 28 the standard reference implementation), and realised-variance frameworks [29, 30] all reintroduce structure at the variance-process layer while preserving the unimodal-innovation constraint. The continuous-latent counterpart is the stochastic-volatility family of Taylor [31], which evolves log-volatility as a latent AR(1) and proceeds by simulation-based estimators [32, 33], hidden-Markov approximations on a discretised volatility grid [34, 35], or Markov-switching SV [36] that interpolates between the continuous-volatility and finite-state formulations. Across both families the recurring failure mode is the same: a single innovation distribution shared across volatility regimes cannot reach the empirical kurtosis target while the variance recursion already constrains the temporal structure, and the regime-switching extensions reintroduce latent states only at the variance-process layer rather than at the marginal-density layer where the present CHMM lives. The structural distinction matters for the regime-conditional Value-at-Risk, because a state-conditional predictive density is available only when the latent state carries an explicit per-state emission family, and the kurtosis match empirically separates a finite-state model with regime-specific heavy-tailed emissions from a single continuous-volatility latent (the kurtosis values in Table S2 and the extended baseline panel make this concrete).

Deep generative market models trade per-state interpretability for high-capacity unconditional fidelity. Generative adversarial network (GAN)-based generators reproduce the marginal appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [37, 38], with the TimeGAN family of Yoon et al. [8], the convolutional Wasserstein GAN QuantGAN of Wiese et al. [9], and the autoregressive-diffusion approach of Rasul et al. [10] the standard references. Path-signature generators [39–42] and score-based or diffusion approaches [10, 43–45] extend the family with finer time-series structure but inherit the same trade-off: tighter visual fidelity to the observed price tape, opaque per-state semantics, no parametric copula handle for multi-asset composition, and no closed-form predictive density for downstream conditional-coverage tests. We included QuantGAN as the deep-generative row in our extended panel, where it functions as a sanity check on whether high-capacity architectures recover the same stylized facts at our sample sizes rather than as a forecasting baseline; we ground our contribution against parametric and non-parametric baselines with the deep row read as a negative control. The recurring failure mode across the deep family is mode collapse on the heavy-tail axis combined with the absence of an explicit state variable, so the per-state semantics that the CHMM uses for regime-conditional Value-at-Risk and copula composition are not available off the shelf, and a deep generator that wins on raw single-window distributional fit still does not afford the use cases this paper exercises.

Cross-asset synthesis, synthetic-data evaluation, and non-stationarity handling round out the literature this paper draws on. The Single Index Model (SIM) [46] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal, and Sklar's theorem [47, 48] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [49, 50] the standard family choices and rank-reordering simulation [51] the standard injection that preserves marginals exactly. Dynamic-conditional-correlation models [52] and composite-likelihood estimators for vast covariance matrices [53] are the standard alternatives at higher dimensions, and our six-asset construction sits well below the dimensions where those estimators become essential while the rank-based copula architecture composes with either at scale. Synthetic-data evaluation in this work is anchored on the framework of Stenger et al. [54], the proper-scoring-rule literature [55–57], kernel two-sample tests [58], and signature-based metrics [39, 59], which together fix the comparison metrics (KS, ACF-MAE, kurtosis, CRPS, signature distance) that the main panel uses. Finally, the walk-forward stress folds are regime introductions that the in-sample distribution does not span, and their natural complements are Bayesian online change-point detection [60] and the particle-filter recursion of Fearnhead and Liu [61], both of which detect the regime introduction rather than absorbing it into a static fit; integrating online detection with the present CHMM is left as a follow-up direction.

3 Methods

3.1 Data and model

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024, a ten-year in-sample window of length $T_{\mathrm{IS}} = 2{,}516$ trading days (IS hereafter), with the period from January 4, 2024 through April 20, 2026 held out for out-of-sample evaluation, $T_{\mathrm{OoS}} = 572$ trading days (OoS hereafter). Prices are split-adjusted but not dividend-adjusted; the data sources are described in the acknowledgments. We checked the 323-day Polygon vs Alpaca / IEX overlap window and found zero volume-weighted average price (VWAP) and return differential and Kolmogorov-Smirnov (KS) $p = 1.00$, so the two vendors agree exactly over the overlap and the stitched series has no visible break at the join. For ticker i and trading day j, with $P_{i,j}$ the session VWAP, $\Delta t = 1/252$ the annualised trading-day step, and r_f the risk-free rate (set to zero here), the annualised excess growth rate is

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0. \tag{1}$$

We use daily data and the matching annualised growth rates, but the framework does not depend on the time step and can be applied to intraday data or other frequencies.

The model is fit one ticker at a time. We write G_t for the daily excess growth rate on trading day t for the ticker being fit (equation (1) with the ticker index dropped and the day index renamed from j to t). The sequence $\{G_t\}$ is the observed output of a continuous hidden Markov model with K latent states. We write the model as $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{F}, \boldsymbol{\pi})$ [62], where $\mathcal{S} = \{1, \ldots, K\}$ is the finite state space and $s_t \in \mathcal{S}$ is the (unobserved) latent state at time t. The transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$ has its (i, j) entry equal to the probability of moving from state i at time t to state j at time $t+1$, $T_{ij} = \mathbb{P}(s_{t+1} = j \mid s_t = i)$; the rows therefore sum to one (each row is a valid probability distribution over next states). The initial-state distribution $\boldsymbol{\pi} \in \mathbb{R}^K$ has kth entry $\pi_k = \mathbb{P}(s_1 = k)$, the probability that the chain starts in state k. The per-state emission family is $\mathbf{F} = \{f_k(\,\cdot\,; \boldsymbol{\theta}_k)\}_{k=1}^K$, with $f_k(x; \boldsymbol{\theta}_k)$ the conditional probability density of the observation G_t given $s_t = k$, evaluated at the real-valued argument x over the support of the daily excess growth rate; the per-state parameter vector $\boldsymbol{\theta}_k$ holds the location, scale, and any shape parameters of the kth emission family. The per-state cumulative distribution function (CDF) is $F_k(x; \boldsymbol{\theta}_k) = \int_{-\infty}^x f_k(u; \boldsymbol{\theta}_k)\, du$, used in the regime-conditional VaR construction.

Let $\bar{\boldsymbol{\pi}} = (\bar{\pi}_1, \ldots, \bar{\pi}_K) \in \mathbb{R}^K$ denote the stationary state distribution, the unique probability distribution over states that is left unchanged by one transition step ($\bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}$), so each component $\bar{\pi}_k$ is the long-run fraction of time the chain spends in state k. Under $\mathcal{M}$, the stationary marginal density of G_t evaluated at x is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k \, f_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}. \tag{2}$$

Existence and uniqueness of $\bar{\boldsymbol{\pi}}$ require that $\mathbf{T}$ be irreducible (every state can eventually be reached from every other state) and aperiodic (returns to a state are not locked to a fixed period); Assumption 1 states the technical conditions. We learn $\mathbf{T}$, the per-state parameter vectors $\boldsymbol{\theta}_{1:K} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K)$, and $\boldsymbol{\pi}$ jointly by expectation-maximisation (EM). The four emission families share a single forward-backward algorithm and quantile-based initialisation, and the M-step is the only difference between variants. CHMM-N uses Gaussian emissions, $f_k = \mathcal{N}(\mu_k, \sigma_k^2)$ with $\sigma_k > 0$. CHMM-t uses Student-t, $f_k = t_{\nu_k}(\mu_k, \sigma_k)$ with $\sigma_k > 0$ and $\nu_k \in [\nu_{\min}, \nu_{\max}]$. CHMM-L uses Laplace, $f_k = \mathrm{Laplace}(\mu_k, \beta_k)$ with $\beta_k > 0$; we write β rather than the canonical b to avoid notational clash. CHMM-GED uses the generalised error distribution, $f_k = \mathrm{GED}(\mu_k, \alpha_k, p_k)$ with $\alpha_k > 0$ and $p_k \in [p_{\min}, p_{\max}]$; CHMM-N and CHMM-L sit at the boundary of CHMM-GED, with $p_k = 2$ recovering Gaussian and $p_k = 1$ recovering Laplace. The CHMM architecture is shown in Fig. 1.

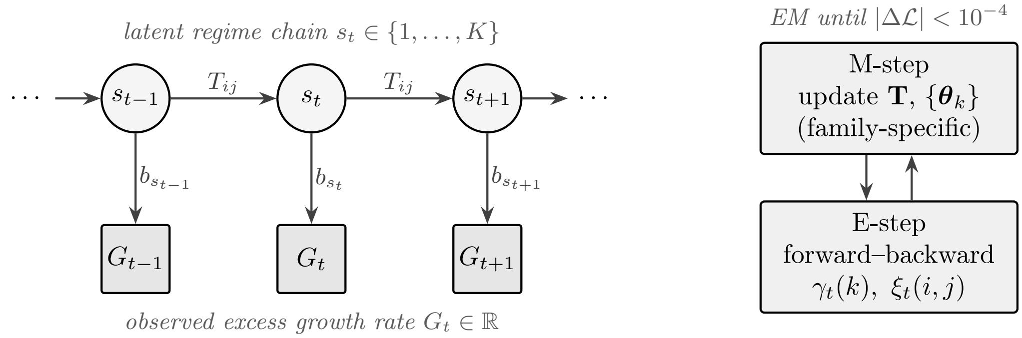

Figure 1. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t emits the observed excess growth rate G_t through a state-specific density b_{s_t}. The four CHMM variants share everything except the per-state emission family (Gaussian, Student-t, Laplace, GED). EM alternates an E-step (log-space forward-backward) and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

3.2 Estimation by EM

Given the model $\mathcal{M}$, we estimate the parameters $(\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi})$ by EM in log-space. Throughout this subsection we write $O_t \equiv G_t$ for the observation at time t (O is the standard symbol for an HMM observation; here it just renames G_t), $O_{1:T} = (O_1, O_2, \ldots, O_T)$ for the full observed sequence over the fitting window of length T, and $\gamma_t(k) = \mathbb{P}(s_t = k \mid O_{1:T})$ for the smoothed posterior probability that the chain was in state k at time t, given the entire observed sequence and the current parameter iterate ("smoothed" means computed using both past and future observations, not just past). The collection $\{\gamma_t(k)\}_{t=1,\ldots,T;\, k=1,\ldots,K}$ is computed at every E-step by the standard forward-backward recursion (one forward pass and one backward pass through the data). We also use the superscript (n) to mark the value of any quantity at EM iteration n, so (0) is the initial value. All four emission families share the same log-space forward-backward recursions [63] and the same quantile-based initialisation: observations are sorted, split into K equal-sized chunks, and state k's initial location and scale are seeded from chunk k, with the transition matrix and initial distribution started uniform, $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. The EM loop runs until the incomplete-data log-likelihood $\mathcal{L}^{(n)} = \log p(O_{1:T} \mid \mathbf{T}^{(n)}, \boldsymbol{\theta}_{1:K}^{(n)}, \boldsymbol{\pi}^{(n)})$ stops changing meaningfully between iterations, $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < \epsilon$, or a maximum iteration cap is reached; specific values for the tolerance, iteration cap, bracket widths, and one-dimensional-search settings are reported in the algorithmic appendix. Quantile-based initialisation is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit. Random initialisations standard in the 1990s often converge to overlapping local optima, which is one of the two combined causes of the low-K failure observed by Rydén et al. [11]; the other is the spectral constraint from the bilinear ACF identity. Algorithmic details and the forward-backward recursion are reported in the appendix.

The four emission families differ only in the M-step: given the smoothed posteriors $\gamma_t(k)$ from the shared E-step, each family has its own update rule for the per-state parameter vector $\boldsymbol{\theta}_k$. For CHMM-N, the per-state parameters are the Gaussian mean and variance, $\boldsymbol{\theta}_k = (\mu_k, \sigma_k^2)$, and the classical Baum-Welch M-step [64] updates them in closed form using weighted sample-mean and sample-variance formulas, where each observation is weighted by the posterior probability $\gamma_t(k)$ that state k generated it (all sums $\sum_t$ below run over $t = 1, \ldots, T$):

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k)\, (O_t - \mu_k)^2}{\sum_t \gamma_t(k)}.$$

For CHMM-t, the per-state parameters are the Student-t location, scale, and degrees of freedom, $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$. We use the expectation conditional maximisation (ECM) formulation of Peel and McLachlan [19] and Liu and Rubin [20], which replaces the single M-step with a sequence of conditional maximisation (CM) sub-updates that each move one block of parameters at a time; this is the same heavy-tailed innovation choice used by Abanto-Valle et al. [35] for stochastic-volatility-in-mean models. The construction uses a standard representation of the Student-t as a Gaussian with random variance: conditional on a latent positive scalar u, the observation is Gaussian, $G_t \mid s_t = k,\, u \sim \mathcal{N}(\mu_k, \sigma_k^2/u)$, and u itself is drawn from a Gamma distribution, $u \sim \Gamma(\nu_k/2,\, \nu_k/2)$. Marginalising over u recovers the Student-t. ECM treats u as an additional latent variable, and the E-step adds its conditional expectation under the current parameter iterate,

$$u_{t,k} \equiv \mathbb{E}[u \mid O_t, s_t = k] = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{3}$$

and the M-step then updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{4}$$

for the location and scale, both in closed form given ν_k. The degrees of freedom ν_k are then updated by a one-dimensional search on the bounded interval $[\nu_{\min}, \nu_{\max}]$ that maximises the state-k piece of the expected complete-data log-likelihood, $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$. Each sub-update is a partial maximisation on a bounded interval, so the standard EM monotonicity guarantee still holds: the log-likelihood of the observed data does not decrease from one iteration to the next.

For CHMM-L, the per-state parameters are the Laplace location and scale, $\boldsymbol{\theta}_k = (\mu_k, \beta_k)$, and both have closed-form updates. The location μ_k is a weighted median of the observations $\{O_t\}_{t=1}^T$, where each observation is weighted by the posterior probability $\gamma_t(k)$ that state k generated it. The scale $\beta_k \leftarrow \sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k)$ is the weighted mean absolute deviation of the observations from μ_k, using the same weights.

For CHMM-GED, the per-state parameters are the GED location, scale, and shape, $\boldsymbol{\theta}_k = (\mu_k, \alpha_k, p_k)$, with conditional density $f_k(x; \mu_k, \alpha_k, p_k) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$. The three parameters cannot all be updated together in closed form, so the M-step is split into three sequential CM updates, each moving one parameter while the others stay fixed. The location is updated by a one-dimensional search on a bounded interval centred on the previous iterate, $\mu_k \leftarrow \arg\min_\mu \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$. The scale then has a closed-form update at the just-updated location, $\alpha_k \leftarrow \left[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\right]^{1/p_k}$, where $W_k = \sum_t \gamma_t(k)$ is the total posterior weight on state k. The shape is updated last by a one-dimensional search on the bounded interval $[p_{\min}, p_{\max}]$, $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k)\, \log f_k(O_t; \mu_k, \alpha_k, p)$, at the just-updated location and scale. As with CHMM-t, each sub-update is a partial maximisation on a bounded interval, so EM monotonicity still holds. CHMM-N and CHMM-L sit at the boundary of this family ($p_k = 2$ for all k gives Gaussian, $p_k = 1$ gives Laplace).

3.3 Evaluation

With the model fitted, we evaluate the CHMM at two scopes across four data universes. The data universes are the SPY in-sample and out-of-sample windows of the data subsection above, a sector-balanced 30-ticker panel spanning ten GICS sectors with three large-cap representatives each, a CRSP cross-decade transfer between a 1994 to 2004 in-sample window and a 2004 to 2006 out-of-sample slice, and a six-asset US-equity basket (SPY, NVDA, JNJ, JPM, AAPL, QQQ).

The *single-asset* evaluation fits one CHMM per ticker and covers the stylized-fact diagnostics, K selection, main generator comparison, cross-ticker generalisation panel, CRSP cross-decade transfer, walk-forward stress test, and Value-at-Risk panel. The *multi-asset* evaluation holds the per-asset CHMM marginals fixed and adds cross-asset dependence on top through a rank-based Student-t copula [49, 50] on the six-asset basket. We select the copula degrees of freedom $\nu^\star$ by profile maximum likelihood and draw joint samples using the rank-reordering simulator of Iman and Conover [51], which preserves each fitted CHMM marginal exactly under Sklar's theorem [47]. A single-index model (SIM), a Gaussian copula on the same CHMM marginals, and a truncated C-vine are reported in the appendix as multi-asset comparators.

The single-asset CHMM is benchmarked against one representative from each generator tier, all fit on the same IS window: a stationary i.i.d. bootstrap [65] as the strongest non-parametric baseline; Gaussian and Laplace i.i.d. generators as marginal baselines; GARCH(1,1) with Gaussian [7] and Student-t [23] innovations as conditional-variance baselines; and Markov-switching GARCH (MS-GARCH) [27] as the closest regime-switching comparator. Extended GARCH-family, semi-Markov, deep-generative QuantGAN [9], stochastic-volatility, multifractal, and jump-diffusion baselines are reported in the appendix. For every model in the panel we simulate P independent paths of length T matching the fitting window [54]; the path count and the deterministic seed used in each run are reported in the corresponding result table. We score the panel on three primary metrics. The first is the two-sample Kolmogorov-Smirnov (KS) pass rate at significance level $\alpha = 0.05$ [66, 67], the fraction of simulated paths whose marginal distribution is not rejected against the observed window; this measures distributional fidelity at the marginal level. The second is the mean simulated excess kurtosis, read against the observed kurtosis on the same window; this measures how well the heavy-tailed marginal is reproduced. The third is the mean absolute error (MAE) between the observed and path-averaged simulated autocorrelation function (ACF) of absolute returns $|G_t|$ over $L = 252$ lags (one trading year),

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)\right|, \tag{5}$$

where τ is the lag index, $|G| = \{|G_t|\}_{t=1}^{T}$ is the absolute-return sequence, $\hat{\rho}^{\text{obs}}_{|G|}(\tau)$ is the sample autocorrelation of $|G|$ at lag τ on the observed window, and $\bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)$ is the same quantity averaged across the P simulated paths; this captures volatility clustering, the third Cont stylized fact. The Value-at-Risk panel additionally reports the unconditional Kupiec coverage test [68], the Christoffersen joint conditional-coverage test [69] as our primary VaR backtest, and an expected-shortfall (ES) envelope. Auxiliary distributional metrics (block-bootstrap KS recalibration, Anderson-Darling (AD), Wasserstein-1 (W_1), Hellinger distance, and continuous ranked probability score (CRPS)) are reported in the appendix to confirm robustness of the cross-generator ranking.

4 Spectral Mechanism

Let $\mathbf{v}_k$ and $\mathbf{w}_k$ denote the right and left eigenvectors of $\mathbf{T}$ at non-unit eigenvalue λ_k, and let $\sigma^2_{|G|}$ denote the stationary variance of $|G_t|$ under the marginal mixture (2). The closed-form bilinear identity for the absolute-return ACF of a stationary CHMM, $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \text{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ with $\mathbf{m}_k = \mathbb{E}[|G_t| \mid s_t = k]$, and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are standard in the regime-switching literature (13, §22.2; 14, Ch. 3; 15). Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1, with $\bar{\boldsymbol{\pi}}$ the unique stationary distribution; 70) and finite per-state second moments with non-trivial location and scale contrast (Assumption 2), the spectral decomposition

$\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^{K} \lambda_k^\tau \, \mathbf{v}_k \mathbf{w}_k^\top$ subtracts the marginal product to give the ACF

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k \, \lambda_k^\tau, \qquad w_k = \big(\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{v}_k\big)\big(\mathbf{w}_k^\top \mathbf{m}\big)/\sigma_{|G|}^2. \tag{6}$$

This is a real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$; complex-conjugate λ_k pairs combine into damped oscillatory modes, and non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks. A self-contained derivation, the lag-zero behaviour, and the squared-return ACF are reported in the appendix, together with the formal Assumptions 1 and 2 and the cited propositions on identifiability, MLE consistency, and ECM monotonicity. Our contribution here is purely empirical. We apply equation (6) to recast the Rydén et al. [11] low-K failure on the SPY 2014 to 2024 instance as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, and we show empirically that the implied $K-1$ rank bound is non-binding at $K \geq 3$.

This identity recasts the Rydén low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$. At $K = 2$, the ACF reduces to the mono-exponential form $\rho_{|G|}(\tau) = \lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$, and slow decay is matched by tuning $T_{11} + T_{22}$. The temporal part is therefore satisfiable at $K = 2$; what binds is the two-component marginal, which cannot match the empirical kurtosis target. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture has enough components to match the observed tail. Empirically, the absolute-return ACF-MAE was nearly flat across $K \in \{3, 6, \ldots, 21\}$ while distributional KS rose, and a direct effective-rank diagnostic showed a single non-unit eigenvalue carrying 93.6% of the lag-1 ACF at $K = 18$ on SPY (Table S5), so the K-rank upper bound is non-binding at $K \geq 3$ on this instance. Repeating the diagnostic on the 30-ticker cross-section gave a wider distribution with cross-ticker median dominant-mode share 0.76 and minimum 0.326 on NEM (Table S6); the rank-non-binding claim is supported at the cross-ticker median rather than as a universal property of equity-return data.

5 Empirical Study

We computed sample stylized-fact diagnostics on the SPY in-sample (IS) and out-of-sample (OoS) windows and observed all three Cont stylized facts on both windows (Figure S3, Table 1 Observed row). The marginal was heavy-tailed (in-sample excess kurtosis 7.68, out-of-sample 5.29), the linear autocorrelation in the raw growth rate G_t was negligible, and the volatility clustering in $|G_t|$ was persistent. A stationary block bootstrap at mean block length $L = 20$ gave broadly overlapping IS and OoS 95% CIs of $[2.17, 12.40]$ and $[0.90, 8.26]$ on the kurtosis point estimates (Table S7), so kurtosis targets should be read against the joint envelope rather than the point estimates. Having confirmed the three stylized facts on both windows, we next swept $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and observed that the $K = 3$ versus $K = 6$ comparison sat inside sampling noise under four-fold and six-fold rolling-origin cross-validation (CV; full K-sweep CV panel in the appendix). The held-out log-likelihood differential satisfied $|z| \leq 0.07$ with the sign flipping between fold designs, and the Bayesian information criterion (BIC) also picked $K = 3$. We chose $K^\star = 3$ as the main-paper value of K: held-out criteria could not distinguish $K = 3$ from $K = 6$, so we picked the smaller model, and larger K was decisively worse on every held-out criterion. Absolute-return ACF-MAE was nearly flat across the whole sweep, consistent with Eq. (6).

With the state count fixed, we fit the four CHMM variants on the SPY in-sample window at $K^\star = 3$ and observed out-of-sample performance using six benchmark generators and an explicit-duration semi-Markov reference (Table 1). The CHMM-N row landed at 89.7% IS and 80.5% OoS, simulated kurtosis 3.83 and 3.53, and $|G_t|$ ACF-MAE 0.0460 (on par with GARCH at 0.0485). The shared-ν Student-t row gave the cleanest heavy-tail match without a penalty hyperparameter at simulated kurtosis 4.68 and 4.46 against observed 7.68 and 5.29, with 91.9% IS and 82.1% OoS KS. The penalised CHMM-t at $\lambda = 20$ is reported as a sensitivity reference: the IS aggregate 14.91 is a $1/\nu_k$-shrinkage outcome with λ tuned at $K = 18$ and re-used at $K^\star = 3$, where the lower-bracket-pinned state carried a larger share of the three-state mixture (discussed below); CHMM-GED's per-state shape landed at 5.48 and 5.12, and CHMM-L at 5.30 and 4.85 (losing roughly 14pp on OoS KS at $K = 3$ against CHMM-GED). The mixture-of-eigenvalues sum (6) reproduced the slow absolute-return ACF that the Rydén et al. [11] $K = 2$ to 3 Gaussian setup could not recover on the SPY 2014 to 2024 instance, and the raw-return ACF-MAE closed the third stylized fact at 0.0235 to 0.0240, indistinguishable from the i.i.d. baselines (Table 1, G_t column). The four CHMM variants were statistically indistinguishable on the OoS sample-CRPS [57] (CHMM-N 1.0393, CHMM-t-pen 1.0373, CHMM-L 1.0405, CHMM-GED 1.0389, against bootstrap 1.0398 and GARCH(1,1) 1.0440), so the family choice was driven by the per-row kurtosis match (Table S2); the cross-generator ranking was robust to the distributional-metric choice across KS, Anderson-Darling, Wasserstein-1, and Hellinger. The MS-GARCH benchmark stayed well below the CHMM on KS under both point-estimate and Bayesian posterior-predictive variants (Table 1; full MS-GARCH panel in the appendix). A six-fold rolling-origin walk-forward (Table S28) reached median OoS KS 62.1% at $K^\star = 3$, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% that every generator in the panel rejected. The walk-forward median, not the single-window OoS pair, is the right summary to use when judging the model for live deployment. The i.i.d. bootstrap and a maximum-likelihood HSMM-N benchmark both exceed the CHMM on raw single-window OoS KS (Table 1); the CHMM differentiates on use cases neither alternative can serve, with the full HSMM panel, state-assignment inspection, bootstrap discussion, and block-bootstrap KS recalibration reported in the appendix.

Having established the main ranking on SPY, we next fit the penalised CHMM-t at $K^\star = 3$ and $\lambda = 20$ on each ticker of a sector-balanced 30-ticker panel (ten GICS sectors with three large-cap representatives each) and observed cross-ticker generalisation behaviour (Table S3). The in-sample distribution was concentrated (median 96.8%) and the out-of-sample distribution was wider, with median 69.1%, mean $66.2 \pm 28.2\%$, and 11/30 tickers below 60%. The deepest failures were driven by tickers that introduced a new regime in the OoS window (LLY and UNH in Health Care, NEM in Materials; full per-ticker rollup at the $K = 18$ kurtosis-match check in Table S4), and the remaining eight sub-60% tickers and the per-sector breakdown at the $K = 18$ check on the kurtosis match (with the same 11/30 failure count and overlapping ticker list) are reported in the appendix. An ANOVA on OoS KS by sector returned no significant sector effect on either the original $n = 3$ design or an $n = 6$ expansion to a 60-ticker panel (full ANOVA panel in the appendix), so the visible concentration of failures on regime-introducing tickers is a per-ticker observation rather than evidence about sector-level structure. We recommend periodic refit: at $K^\star = 3$ a quarterly refit shifted the OoS KS median from 69.1% to 84.7% and reduced failures from 11/30 to 8/30, while the same cadence at the $K = 18$ kurtosis-match check gave median 83.0% and 7/30, with monthly cadence interpolating to 86.7% and 5/30 at three times the compute (Table S33). Sensitivity panels at $K = 6$ and $K = 18$ and the SPY-level shared-ν Student-t alternative are reported in the appendix.

We then composed per-asset CHMM-N marginals at $K^\star = 3$ through a rank-based Student-t copula on a six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) and observed off-diagonal correlation reproduction across the four candidate dependence layers (Table 2; full comparator panel in the appendix). On the in-sample window the Student-t copula gave off-diagonal correlation MAE of 0.027 over 200 simulated paths, against 0.030 for the Gaussian copula, 0.068 for the truncated C-vine, and 0.076 for the single-index model (SIM). Profile MLE selected $\nu^\star = 6$ on the unit-spaced grid $\nu \in [4, 12]$ with Wilks 95% profile-LL CI of $[6, 7]$ on the IS slice, and a full one-shot MLE jointly maximising over (Σ, ν) confirmed the choice was robust to the two-step estimator. The off-diagonal MAE was essentially K-robust between $K^\star = 3$ and the $K = 18$ kurtosis-match

Table 1. Main generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS $T = 2{,}516$, OoS $T = 572$. Columns: KS = Kolmogorov-Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute-return / raw-return ACF over 252 lags. Bold marks the best entry on each axis *within block* (block = i.i.d. baselines, GARCH family, MS-GARCH, CHMM, co-main HSMM). The 1-day OoS KS column is dominated by the i.i.d. bootstrap; the CHMM differs from the bootstrap on use cases the bootstrap cannot serve (regime-conditional VaR, multi-asset copula composition), not on per-day distributional fidelity. Sample-CRPS is reported as a tie-break in the prose paragraph following the table; the four CHMM rows are statistically indistinguishable on this metric. Block-bootstrap KS recalibration, the extended GARCH-family and MS-GARCH-$K \in \{3, 4, 6\}$ panel, and the $K = 6$ and $K = 18$ sensitivity panels are reported in the appendix. ¶ Reference Bayesian re-run of MS-GARCH via the `MSGARCH` R package of Ardia et al. [28]; the lower KS pass rate reflects posterior-predictive integration of parameter uncertainty. § QuantGAN row is an in-house WGAN re-implementation (3-layer 1D-conv generator and critic, Wasserstein loss; full spec in the appendix); the row reads as the panel's deep-generative negative control. §§ Shared-ν Student-t ablation: single ν across all K states by aggregate-Q ECM, no penalty.

	KS (%) ↑		Exc. Kurt		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t \rvert$	G_t
Observed	–	–	7.68	5.29	–	–
Bootstrap	**99.7**	**92.1**	**7.24**	**6.81**	0.0628	**0.0235**
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243
GARCH(1,1)-t	**57.3**	**80.8**	15.13	–	**0.0316**	**0.0173**
MS-GARCH ($K = 2$)	27.7	**38.7**	**4.73**	–	0.0367	0.0173
MS-GARCH ($K = 3$)	**36.1**	33.1	4.10	–	**0.0284**	0.0173
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.0429	0.0173
MS-GARCH ref. Bayesian ($K = 2$)¶	0.0	5.8	4.00	2.46	0.0465	0.0240
MS-GARCH ref. Bayesian ($K = 3$)¶	0.1	5.1	4.52	2.53	0.0433	0.0241
MS-GARCH ref. Bayesian ($K = 4$)¶	0.0	5.3	6.01	3.54	0.0446	0.0241
QuantGAN TCN§	0.0	0.0	0.56	0.53	0.0617	0.0264
Main-paper choice: $K^\star = 3$ (selected by held-out criteria under rolling-origin CV).						
CHMM-N	89.7	80.5	3.83	3.53	**0.0460**	0.0240
CHMM-t pen. ($\lambda = 20$)	90.6	**83.2**	14.91	8.50	0.0537	0.0235
CHMM-L	79.8	63.1	5.30	4.85	0.0532	0.0236
CHMM-GED	90.5	77.4	5.48	5.12	0.0526	0.0236
CHMM-t shared-ν§§	**91.9**	82.1	**4.68**	**4.46**	0.0531	**0.0235**
ML HSMM-N	**98.4**	**91.0**	**3.46**	**3.38**	**0.0629**	**0.0236**

check (Table 2 caption). The universe contained overlapping ETFs and constituents (QQQ holds AAPL and NVDA; SPY holds all four single names), so the dependence structure was dominated by strong positive co-movement and the copula task was easier than on a non-overlapping basket; a non-overlapping comparison panel is reported in the appendix. On OoS the Student-t versus Gaussian gap fell within path-to-path noise at this N_{paths}: the off-diagonal MAE was 0.209 for Student-t and 0.202 for Gaussian, a 0.007 gap that we report descriptively rather than under a formal paired test (Table 2). Out-of-sample, the choice that matters is the quarterly-rolling refit, which reduced the OoS off-diagonal MAE from 0.209 to 0.185 (Table 2); the refit-cadence choice dominates the dependence-family choice out-of-sample. The cross-asset OoS distribution was bimodal: NVDA and JPM failed OoS KS while the other four passed above 77%, driven by single-name regime shifts (NVDA on the 2024-2025 AI rally, JPM on the 2023 regional-bank stress), and the failure count recovered to 0/6 under quarterly refit (Table 2; full rolling-copula panel in the appendix).

We back-tested a regime-conditional VaR built from the CHMM one-step-ahead state forecast

Table 2. Cross-asset Student-t copula on CHMM-N marginals at $K^{\star} = 3$ (multi-asset construction; $\nu^{\star} = 6$; 200 paths; seed `20260422`). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction averaged over paths. Main-paper $K^{\star} = 3$ matches the single-asset Table 1. The off-diagonal MAE is essentially K-robust (the same panel at the $K = 18$ kurtosis-match check gives 0.027/0.209, identical to two decimal places); per-asset KS sits lower at $K^{\star} = 3$ than at $K = 18$, but the cross-asset use case is dominated by the dependence layer rather than per-asset marginal fit. The comparison against SIM, Gaussian, and truncated C-vine is reported in the appendix.

Ticker	KS (%) ↑ IS	KS (%) ↑ OoS
SPY	87.0	77.5
NVDA	92.0	62.5
JNJ	97.5	94.0
JPM	97.5	57.5
AAPL	85.0	86.0
QQQ	83.0	85.5
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Off-diag MAE OoS (quarterly refit, $K = 18$ reference)	**0.185**	

(construction in Methods) under the Christoffersen joint conditional-coverage and Engle-Manganelli dynamic-quantile (DQ) tests on SPY OoS, and observed clean passes at $\alpha = 0.05$ across all four emission families at both $K \in \{3, 18\}$ (Table 3). The four CHMM rows passed Christoffersen-cc at $\alpha = 0.05$ with $p_{\mathrm{cc}} \in [0.49, 0.68]$, and CHMM-N at $K^{\star} = 3$ also passed the DQ test ($p = 0.16$). Against two non-state-space conditional-VaR baselines (filtered bootstrap [71] and the Engle-Manganelli symmetric-absolute-value CAViaR specification [72]) the CHMM rows reached approximate parity at $\alpha = 0.05$ and a strict-tail advantage for CHMM-N at $K^{\star} = 3$ at $\alpha = 0.01$: only CHMM-N at $K^{\star} = 3$ avoided DQ rejection at $\alpha = 0.01$, while the filtered bootstrap and CAViaR contenders both rejected. Extending to six rolling-origin walk-forward folds, the Christoffersen-cc test passed on 19/24 rows at $\alpha = 0.05$, with failures concentrated on the W2 (COVID 2020) and W4 (2022 rate-hike onset) folds that every generator in the panel rejected (full walk-forward and refit-cadence panels, four-family ablation, and Benjamini-Hochberg false-discovery-rate correction are reported in the appendix).

The four independent stress sources tested above (cross-ticker generalisation, cross-decade transfer, the GLD non-equity stress, and the walk-forward W2 and W4 folds) all reduce to the same pattern: static IS fits collapse on slices whose marginal moves outside the IS regime library, periodic refit closes the gap on slices that stay inside the equity-return scope, and stress folds where the regime introduction itself falls inside the refit window remain uncovered at every cadence (Table S33). We recommend periodic refit for live use at a cadence chosen against the OoS slice rather than against the IS fit alone.

Table 3. Regime-conditional VaR back-test on SPY OoS ($T_{\mathrm{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The CHMM rows use the forward filter under IS-fixed parameters $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$; only the one-step-ahead state-probability vector updates through OoS. The contender rows (filtered bootstrap and CAViaR) are scored on the same Christoffersen / DQ harness. Every CHMM row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$. ‡ $\alpha = 0.01$ rows are power-bounded at $T_{\mathrm{OoS}} = 572$; the DQ test (higher power) rejects $K = 18$ at $\alpha = 0.01$ ($p = 0.017$, bold in p_{DQ}) and the contender rows also reject at $\alpha = 0.01$ on DQ. The full power calibration is in the appendix. * DQ p-value at 4-lag specification; CHMM-t penalised rows are reported with '–' in this column.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}	$p_{\mathrm{DQ}}{}^*$
CHMM-N	3	$0.01^{\ddagger}$	9	1.57%	−4.56	1.62	2.36	3.98	0.14	0.06
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42	0.49	0.16
CHMM-N	18	$0.01^{\ddagger}$	9	1.57%	−5.20	1.62	2.36	3.98	0.14	**0.02**
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78	0.68	0.68
CHMM-t (λ=20)	3	$0.01^{\ddagger}$	8	1.40%	−5.59	0.82	2.81	3.63	0.16	–
CHMM-t (λ=20)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19	0.55	–
CHMM-t (λ=20)	18	$0.01^{\ddagger}$	10	1.75%	−6.25	2.65	1.98	4.62	0.10	–
CHMM-t (λ=20)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40	0.50	–
Non-state-space conditional VaR contenders.										
Filtered bootstrap	–	$0.01^{\ddagger}$	8	1.40%	−4.81	0.82	2.81	3.63	0.16	**0.04**
Filtered bootstrap	–	0.05	24	4.20%	−2.89	0.82	0.84	1.67	0.43	0.55
CAViaR (SAV)	–	$0.01^{\ddagger}$	6	1.05%	−4.82	0.01	3.97	3.98	0.14	**0.01**
CAViaR (SAV)	–	0.05	25	4.37%	−2.99	0.50	0.67	1.17	0.56	0.76

6 Discussion

The main empirical finding is that a continuous HMM trained by Baum-Welch with quantile-based initialisation reproduces the three symmetric Cont stylized facts on daily US-equity returns at $K^\star = 3$, and the remaining kurtosis gap is set by which heavy-tailed emission family we pick. Two issues combine in the Rydén et al. [11] low-K failure: a single self-transition rate gives the absolute-return ACF a single exponential decay mode, and the random initialisations standard in the 1990s often converged to degenerate local optima. Our $K = 2$ replication on SPY confirmed that what limits the fit is the marginal distribution, not the ACF: at $K = 2$ the KS pass rate dropped sharply while the absolute-return ACF-MAE stayed essentially constant. With modern initialisation the ACF was reproduced at any $K \geq 2$, and the choice of K was then driven by distributional fidelity. The mechanism behind the residual kurtosis gap is simple: CHMM-N at $K^\star = 3$ produced simulated IS kurtosis 3.83 against observed 7.68 (Table 1) because a mixture of only three Gaussians cannot reach the observed heavy tail no matter how the state weights or means are tuned. Switching to a heavy-tailed emission closed most of the gap. The shared-ν Student-t ablation, our primary heavy-tailed specification at $K^\star = 3$, reached simulated IS kurtosis 4.68 with no penalty hyperparameter (Table 1). The penalised per-state CHMM-t at $\lambda = 20$ overshot to IS kurtosis 14.91, with λ tuned at $K = 18$ and re-used at $K^\star = 3$ as an upper bound. CHMM-L sat between CHMM-N and CHMM-t at IS kurtosis 5.30 without any shape parameter. CHMM-GED's bimodal $\hat{p}_k$ partition, with states splitting into a Gaussian-like bulk and a Laplace-like tail, is the data-driven version of the Gaussian/Laplace hybrid one would otherwise assemble by hand-classifying states. The bottleneck at $K \geq 3$ is therefore the marginal mixture's ability to reach the observed kurtosis, not the Markov chain's ability to reproduce the slow ACF; the emission-family ablation tells us which symmetric heavy-tailed family carries the residual gap most clearly.

One real limitation: a static-IS-fit CHMM does not generalise across regime introductions. The

GLD non-equity stress and the W2 (COVID 2020) and W4 (2022 rate-hike) walk-forward folds were regime introductions whose OoS marginal sat outside the range covered by any of the IS states; on those slices, OoS performance broke down (Table S33). Single-name equity returns are well known to carry structural breaks [73–75], and the cross-ticker concentration of failures on names that introduced a new regime (LLY, UNH, NEM) is the same pattern at the panel level. We recommend periodic refit [76, 77] for non-stress equity slices. It does not save GLD or the W2 and W4 stress folds, where the IS distribution simply does not span the OoS slice, and where faster refit cadences also do not close the gap. The implication for K-selection is that the choice should be driven by the marginal distribution, not the ACF. The ACF was reproduced at any $K \geq 2$ once initialisation was fixed, so held-out KS, log-likelihood, and BIC are the criteria that actually move with K on held-out data, while stress-fold rejections diagnose OoS-slice composition rather than IS-axis misspecification.

Beyond the deployment limit just discussed, two further scope limits remain: the stylized facts we evaluated against, and the model size at which we compete with deep generators. The Cont [5] stylized-fact list also includes the leverage effect, the negative $\mathrm{Corr}(G_t, |G_{t+1}|)$ characteristic of equity returns; we evaluated against only the three symmetric stylized facts and leave the leverage axis to future work, since symmetric emissions cannot reproduce a negative cross-correlation by construction. On parametric complexity, CHMM-N at $K = 18$ has $K(K-1) + 2K = 342$ free parameters (CHMM-t adds K degrees-of-freedom parameters, CHMM-GED adds K shape parameters), whereas the convolutional generator-critic architecture of Wiese et al. [9] sits two to three orders of magnitude higher. The CHMM therefore competes against deep generators at a parameter budget small enough to support identifiability analysis, posterior diagnostics, and the closed-form eigenvalue argument that motivates our $K^\star$ choice. The same budget also keeps each per-state emission's location, scale, and shape directly inspectable as an economic regime, whereas a deep generator's latent representation does not factor along regime-level axes. These limits motivate four independent extensions: asymmetric emissions for the leverage effect, a time-varying transition matrix for non-stationarity, factor or vine copula structures for higher dimension, and embedding the model in a portfolio-optimisation or tail-risk-budgeting loop. Each leaves the unified four-family framework intact.

7 Conclusion

A continuous HMM trained by Baum-Welch with quantile-based initialisation reproduces the three symmetric Cont stylized facts on daily US-equity returns. The standard mixture-of-eigenvalues identity for the absolute-return ACF [13–15] recasts the Rydén et al. [11] low-K failure as a rank condition on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$; that condition is not binding at the cross-ticker median once $K \geq 3$, and the bottleneck moves from the Markov chain to the marginal mixture. The unified ECM framework across four emission families isolates the M-step as the only difference between variants and gives a clean side-by-side ablation. The shared-ν Student-t variant narrowed the kurtosis gap at $K^\star = 3$ to 4.68 against observed 7.68 without a tuning hyperparameter (Table 1); CHMM-L sat between Gaussian and Student-t without any shape parameter; and CHMM-GED's bimodal $\hat{p}_k$ partition, with states splitting into a Gaussian-like bulk and a Laplace-like tail, is the data-driven version of the Gaussian/Laplace hybrid one would otherwise assemble by hand-classifying states. The regime-conditional Value-at-Risk, built by averaging the family-appropriate predictive density over the one-step-ahead state forecast, passed the Christoffersen joint conditional-coverage test at $\alpha = 0.05$ on the main OoS window and on 19/24 walk-forward rows (Tables S29 and S30), and the strict-tail Engle-Manganelli test favoured CHMM-N at $K^\star = 3$ over filtered-bootstrap and CAViaR comparators (Table 3). Rejections concentrated on stress folds where the regime moved outside the training distribution, and every generator in the panel rejected on those folds. The main result is bounded in scope. The i.i.d. bootstrap and a maximum-likelihood HSMM-N benchmark beat the CHMM on raw single-window OoS Kolmogorov-Smirnov, and the CHMM is useful for things those alternatives cannot do: regime-conditional VaR, multi-asset copula composition on a sector-balanced 30-ticker panel, and parametric synthetic-data privacy. The dataset scope is daily US equities under stationary OoS (Table S33); the GLD non-equity stress and the W2 and W4 walk-forward folds collapsed under static fitting, and we recommend periodic refit for production use.

These results motivate four extensions. Asymmetric or skewed emission families would extend our coverage of the symmetric stylized facts to the leverage effect of the Cont [5] list, adding a shape parameter that targets the third moment rather than the fourth and complementing the per-state p_k ablation of CHMM-GED. A time-varying transition matrix, estimated by rolling refit, particle filtering, or Bayesian online updates [77], would address the W2 (COVID 2020) and W4 (2022 rate-hike onset) stress folds and the 3/24 persistent walk-forward rejections that the static IS fit leaves open (Table S30). Scaling the multi-asset copula composition beyond the six-asset and 30-ticker panels through factor or vine structures would extend per-asset distributional fidelity to portfolio-level synthesis; the multi-asset construction already separates the per-asset marginals from the dependence layer, so either scaling path slots in without rewriting the CHMM side. Embedding the CHMM-mixed predictive in a portfolio-optimisation or tail-risk-budgeting loop would give an end-to-end test of the regime-conditional VaR as a live trading signal, with the $K^\star = 3$ versus $K = 18$ trade-off (CHMM-N OoS KS 80.5% versus walk-forward median 67.7%; Tables 1 and S28) as the natural sensitivity check. Each extension targets one limitation of the present empirical scope, and the unified ECM framework absorbs all four without structural change.

Conflict of Interest Statement. The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions. J.V. directed the study. A.A. developed the continuous model and simulation code, implemented Baum-Welch, conducted the in-sample and out-of-sample analysis, and generated all figures and tables. C.J. contributed to ECM-derivation methodology review and to validation of the cross-decade and cross-ticker pipelines. All authors edited and reviewed the final manuscript.

Data Availability Statement. The simulation scripts, training and testing data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: `https://github.com/varnerlab/CHMM-paper`. The core model logic (fitting, simulation, decoding, and validation) is implemented in the companion Julia package `CHMM-Model.jl`: `https://github.com/varnerlab/CHMM-Model`. Daily prices are sourced from Polygon.io via Massive (2014-01-03 to 2025-11-18) and Alpaca / IEX (extension through 2026-04-20).

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, and the validation metric definitions. It also reports the formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

Forward-backward recursions and M-step updates. For reference, the in-text Baum-Welch description of the Methods is underpinned by the standard log-space recursions [62, 63]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log f_k(O_1), \tag{7}$$

$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log f_j(O_t), \quad t = 2, \ldots, T, \tag{8}$$

$$\log \beta_T(k) = 0, \tag{9}$$

$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log f_j(O_{t+1}) + \log \beta_{t+1}(j))\,, \quad t = T-1, \ldots, 1, \tag{10}$$

where $\operatorname{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, f_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, f_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{11}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{12}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $f_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [19], Liu and Rubin [20] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{13}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{14}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{15}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [19], Liu and Rubin [20] ECM; the rate sweep and its effect on simulated IS kurtosis are in the body Discussion).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $f_k(x) = \frac{1}{2\beta_k} \exp(-|x - \mu_k|/\beta_k)$ (with the per-state Laplace scale denoted β_k rather than the canonical Laplace-scale symbol b) the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k)), \qquad \beta_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{16}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $f_k(x) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp\big(-(|x-\mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k,\, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k) \left|O_t - \mu\right|^{p_k^{(n)}}, \tag{17}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k) \left|O_t - \mu_k^{(n+1)}\right|^{p_k^{(n)}}\right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{18}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \arg \max_{p \in [p_{\min},\, p_{\max}]} \sum_t \gamma_t(k) \log f_k\big(O_t;\, \mu_k^{(n+1)},\, \alpha_k^{(n+1)},\, p\big), \tag{19}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. This update is the unique zero of $\partial Q_k / \partial \alpha$, so CM-2 is exact maximisation. CM-1 and CM-3 are bracketed maximisation. Each CM step is therefore a partial maximisation, and the standard ECM monotonicity result [78] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

Algorithm descriptions. Algorithm 1 is the unified EM procedure shared across all four emission families: a shared scaffold (quantile-based initialisation, log-space forward-backward, transition update) plus four family-specific branches at lines 3, 13, and 30. Algorithm 2 is the simulator. Algorithm 3 is the cross-asset rank-reordering simulator for the multi-asset construction. Viterbi and SIM-resampling pseudocode are provided in the companion code repository; both are textbook constructions and we omit them from the appendix.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q-function. Hyperparameter defaults used throughout this paper: convergence tolerance $\mathrm{tol} = 10^{-4}$ on $\Delta\mathcal{L}$, `max_iter` $= 60$, 40-iteration golden-section sub-step on every bracketed CM, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 10$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and μ_k search interval $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \mathrm{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.
1: **Quantile-Based Initialization.**
2: Sort $O^{(\mathrm{sort})} \leftarrow \mathrm{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:     **switch** F
5:         $F = \mathcal{N}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$.
6:         $F = t$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad \nu_k \leftarrow \nu^{(0)}$.
7:         $F = L$: $\mu_k \leftarrow \mathrm{median}(C_k)$, $\quad b_k \leftarrow \max(\mathrm{mean}(|C_k - \mu_k|), 10^{-6})$.

$F = \text{GED}$: $\mu_k \leftarrow \text{mean}(C_k)$, $\quad \alpha_k \leftarrow \max(\text{std}(C_k), 10^{-6})$, $\quad p_k \leftarrow p^{(0)}$.
end for
$T_{ij} \leftarrow 1/K$; $\quad \pi_k \leftarrow 1/K$; $\quad \mathcal{L}_{\text{prev}} \leftarrow -\infty$.
EM Loop.
for $n = 1$ to `max_iter` **do**
  E-Step, per-family emission log-likelihood.
  switch F
    $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
    $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
    $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
    $F = \text{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log \Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
  E-Step, shared log-space forward-backward.
  $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
  for $t = 2$ to T, $j = 1$ to K **do**
    $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
  end for
  $\log \beta_T(k) \leftarrow 0$.
  for $t = T - 1$ down to 1, $i = 1$ to K **do**
    $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
  end for
  Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (11).
  Shared transition update. $T_{ij} \leftarrow \sum_t \xi_t(i,j) \;/\; \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
  M-Step, per-family emission update.
  switch F
    $F = \mathcal{N}$ {classical Baum-Welch [64].}
      $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
      $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
    $F = t$ {ECM of Peel and McLachlan [19], Liu and Rubin [20]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
      $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
      $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
      $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
      $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
    $F = L$ {closed-form weighted Laplace MLE.}
      $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
      $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
    $F = \text{GED}$ {three-stage ECM: bracketed μ_k, closed-form α_k, bracketed p_k.}
      $\mu_k \leftarrow \arg\min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40-iter golden-section search.}
      $\alpha_k \leftarrow \max\Big(\big[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\big]^{1/p_k},\ 10^{-6}\Big)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k / \partial \alpha$.}
      $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \text{GED}(O_t; \mu_k, \alpha_k, p)$ {40-iter golden-section search.}
  Convergence check. $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$.
  if $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$ **then**
    break
  end if
  $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$.
end for
return $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$; (μ_k, α_k, p_k) for $F = \text{GED}$.}

Per-state shape diagnostics: CHMM-t and CHMM-GED. This appendix backs the discussion in the body Discussion of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

Algorithm 2 CHMM simulation of synthetic excess growth rate paths (unified four-family scaffold). The emission draw at line 4 branches on the family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$ exactly as in Algorithm 1: $\mathcal{N}(\mu_k, \sigma_k^2)$ for CHMM-N, $t_{\nu_k}(\mu_k, \sigma_k)$ for CHMM-t, Laplace(μ_k, β_k) for CHMM-L, GED(μ_k, α_k, p_k) for CHMM-GED; the latent chain and transition step are shared.

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}, F)$ with $\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, β_k) for $F = L$; (μ_k, α_k, p_k) for $F = \mathrm{GED}$. Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$
1: **for** $p = 1$ to P **do**
2: Sample initial state $S_1 \sim \mathrm{Categorical}(\boldsymbol{\pi})$.
3: **for** $t = 1$ to M **do**
4: Emit observation $\hat{G}^{(p)}_t \sim f_{S_t}(\,\cdot\,; \boldsymbol{\theta}_{S_t})$ with f_k the F-family per-state density.
5: **if** $t < M$ **then**
6: Transition to next state $S_{t+1} \sim \mathrm{Categorical}(\mathbf{T}_{S_t,\cdot})$.
7: **end if**
8: **end for**
9: **end for**
10: **return** $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

Algorithm 3 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^{d}$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\mathrm{Gaussian}, t_\nu\}$, path length T, number of paths P
Ensure: Cross-asset path tensor $\{\hat{G}^{(p)}_{j,t}\}$ of shape $T \times d \times P$
for $p = 1$ to P **do**
Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T \times d}$ from $\mathcal{C}(\Sigma)$:
$L \leftarrow \mathrm{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
$Y \leftarrow Z L^\top$
if $\mathcal{C} =$ Gaussian **then**
$\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
else
Draw $W \sim \chi^2_\nu/\nu$ i.i.d. for $t = 1, \ldots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
$\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
end if
for $j = 1$ to d **do**
Simulate $\tilde{g}^{(p)}_{j,1:T}$ from $\mathcal{M}_j$ via Algorithm 2.
Compute order statistics $\tilde{g}^{(p)}_{j,(1)} \leq \cdots \leq \tilde{g}^{(p)}_{j,(T)}$.
Compute ranks $r^{(p)}_{j,t} \leftarrow \mathrm{rank}(U^{(p)}_{j,t})$ for $t = 1, \ldots, T$.
$\hat{G}^{(p)}_{j,t} \leftarrow \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ for $t = 1, \ldots, T$.
end for
end for
return $\{\hat{G}^{(p)}_{j,t}\}$

CHMM-t per-state ν_k. Figure S1 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

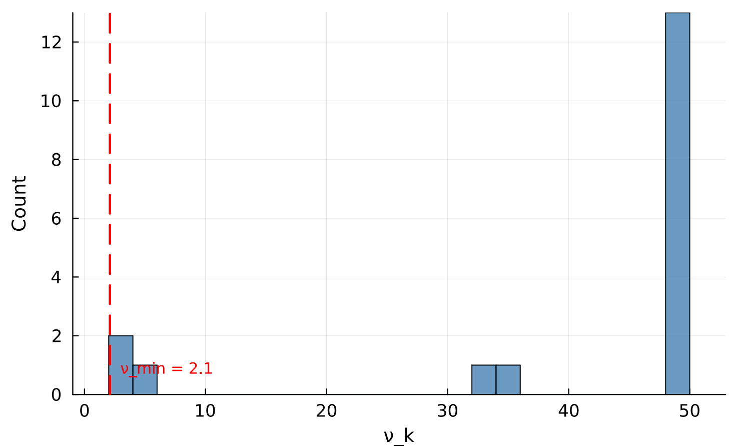


Figure S1. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The body generator comparison reports and discusses the $\hat{p}_k$ partition (Figure S4). Briefly, the $K = 18$ CHMM-GED fit on SPY IS is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$, one state lands at intermediate $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket is non-binding. The bimodal partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (full robustness panel in the cross-asset robustness appendix).

Bracket-sensitivity sweeps. A CHMM-t bracket sweep on $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0, \infty\}$ at $K = 18$ shows the simulated IS kurtosis stays in the 12.5–15.1 band across finite brackets and only collapses to 5.03 in the full Gaussian limit; the overshoot is a feature of the Student-t emission at $K = 18$, not a lower-bracket artefact. The analogous CHMM-GED sweep on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the Gaussian-like-state count invariant at 13/18 and the heavy-shape count in $\{4, 5\}$; tightening $p_{\max}$ from 4.0 to 3.0 costs about 4 nats of LL. The $p_{\min}$ sweep on $\{0.5, 0.7, 0.85\}$ is fully degenerate: every floor produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86.

Evaluation summary. For the single-asset analysis, observed returns G_t are fit by Baum-Welch CHMM at the body operating point $K^\star = 3$, with $K = 18$ retained as the kurtosis-fidelity sensitivity reference; 1,000 paths are then simulated and per-asset metrics computed (Table 1). For the multi-asset analysis we reuse the per-asset $K^\star = 3$ fits as marginals, draw a copula sample $\mathbf{U} \sim \mathcal{C}(\Sigma)$ from the fitted Student-t copula on ranks, rank-reorder the per-asset single-asset paths to match $\mathbf{U}$, and report cross-asset metrics (Table 2). The single-asset CHMM scaffold is shared across both; the CHMM training and simulation steps are Algorithms 1 and 2, the cross-asset rank-reordering step is Algorithm 3, and the architecture diagram for the single-asset scaffold is Figure 1.

A.2 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in the body evaluation protocol. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [66, 67] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{20}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{21}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [79],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| \, du. \tag{22}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{B} \left(\sqrt{p_i} - \sqrt{q_i}\right)^2}, \tag{23}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of the body cross-asset analysis), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} \;=\; \frac{1}{P}\sum_{p=1}^{P}\frac{2}{d(d-1)}\sum_{1\le i<j\le d}\left|\hat{\Sigma}^{(p)}_{\text{sim},\,ij} - \hat{\Sigma}_{\text{obs},\,ij}\right|.$$

A.2.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 1 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

$$\widehat{\text{CRPS}}_t = \frac{1}{N}\sum_{i=1}^{N}|x_i - y_t| - \frac{1}{N(N-1)}\sum_{1\le i<j\le N}|x_i - x_j|. \tag{24}$$

The double sum is computed in $O(N\log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout the body Results; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test on the per-t loss differential $d_t = \widehat{\text{CRPS}}_t^{A} - \widehat{\text{CRPS}}_t^{B}$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3}\rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The body claim of the body generator comparison is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.2.2 Christoffersen Conditional-Coverage Panel (Companion to the body VaR back-test)

The body the body VaR back-test reports the unconditional Kupiec LR_{uc} statistic. This appendix reports the full Christoffersen [69] panel: LR_{uc} (unconditional coverage), LR_{ind} (breach independence), and $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α-quantile of vec(sim archive) across paths and time), then $\text{br}_t = (R_t \le \widehat{\text{VaR}}_\alpha)$.

The headline reads as follows. On the unconditional (Kupiec) leg every CHMM variant passes at both levels on both windows; the LR_{uc} statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clusters at 3.03–3.83 (just below the χ^2_1 critical value 3.841), reflecting the integer-breach grid at $T_{\text{OoS}} = 572$ noted in the body table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejects: LR_{ind} is in the range 4.71–56.46

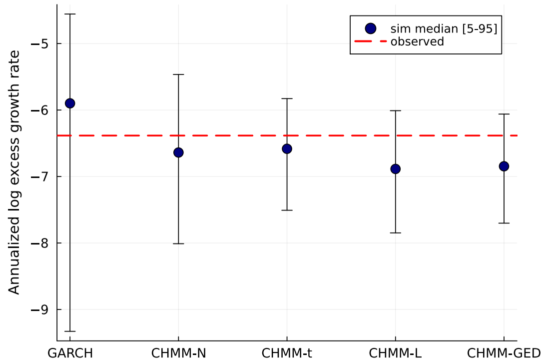
Table S1. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. "br rate" is the empirical breach rate; $\mathrm{LR_{uc}}$ is the Kupiec statistic ($\sim \chi^2_1$ under correct unconditional coverage); $\mathrm{LR_{ind}}$ is the Christoffersen independence statistic ($\sim \chi^2_1$ under independent breaches); $\mathrm{LR_{cc}} = \mathrm{LR_{uc}} + \mathrm{LR_{ind}}$ is the joint conditional-coverage statistic ($\sim \chi^2_2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

					Kupiec		Christ. ind		Christ. cc	
Model	Win	α	breaches	br rate	$\mathrm{LR_{uc}}$	p	$\mathrm{LR_{ind}}$	p	$\mathrm{LR_{cc}}$	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

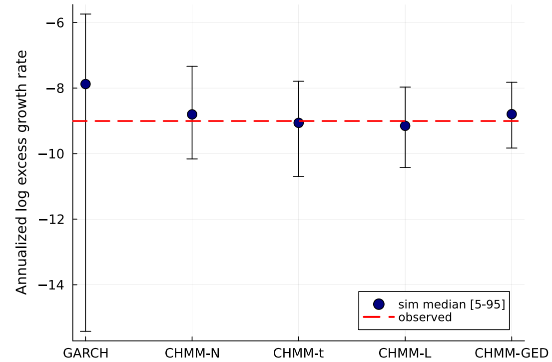
across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility, the OoS-window breaches concentrate inside the high-volatility regime, and a constant-across-t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} \mid \mathrm{history}_t)$ through the per-state emission-mixture; this is operationally the conditional-VaR companion-paper direction noted in the body Discussion.

A.2.3 VaR / ES Envelope Visualization (Companion to Table S14)

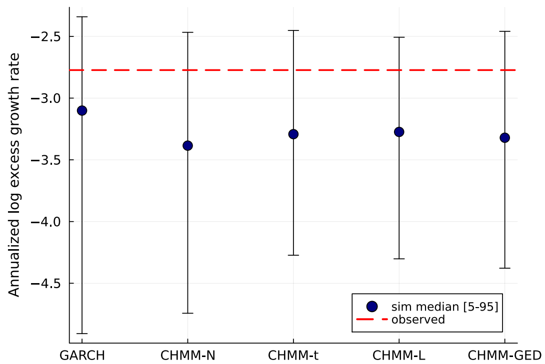
The median and $[5\%, 95\%]$ envelopes across simulated paths are visualised in Fig. S2; the numeric values are reported in Table S14.



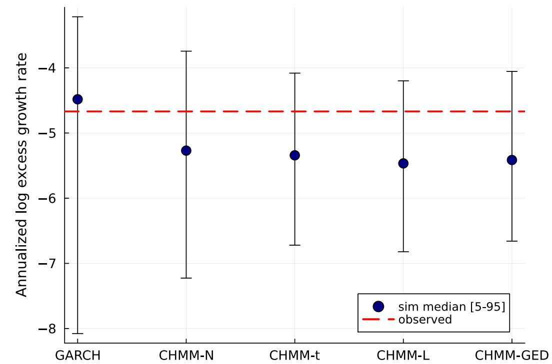
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure S2. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy in the body evaluation protocol). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in the body VaR back-test.

A.3 Formal Theoretical Statements

The following assumptions and propositions are referenced in the body spectral mechanism and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$ is irreducible and aperiodic on the state space $\{1, \dots, K\}$, so the chain admits a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$ and $\lambda_1 = 1$ is a simple eigenvalue with all other eigenvalues strictly inside the unit disk [70].*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density $f_k(\cdot; \boldsymbol{\theta}_k)$ has finite second moment, $\mathbb{E}_{f_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k.*

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L))**.** *Under compactness of the parameter space (μ_k, σ_k or β_k or α_k, $\nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of the body Methods (closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k, followed by a golden-section maximisation of the per-state Q-function over ν_k) satisfies $Q(\theta^{(n+1)} \mid \theta^{(n)}) \geq Q(\theta^{(n)} \mid \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed μ_k, closed-form α_k, bracketed p_k) of equations (17)–(19).*

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [78, 80] converts Q-monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Numerical full-rank check on $\hat{\mathbf{T}}$. The Allman-Matias-Rhodes 2009 identifiability theorem requires $\hat{\mathbf{T}}$ of full rank. The fitted $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS is full rank in the strict sense (smallest singular value $\sigma_{\min} \in [0.0007, 0.0170]$, condition numbers 63–1,620), and the deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has numerical rank exactly $K - 1 = 17$ across all four families.

Proposition 2 (Identifiability up to label permutation)**.** *Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1), the emission contrast condition (Assumption 2), and pairwise distinctness of the per-state parameter tuples $\boldsymbol{\theta}_k$ across $k = 1, \ldots, K$, the CHMM parameter is identifiable up to label permutation for CHMM-N (Gaussian) and CHMM-L (Laplace). For CHMM-t (with per-state $\nu_k \in [\nu_{\min}, \nu_{\max}]$) and CHMM-GED (with per-state $p_k \in [p_{\min}, p_{\max}]$) the same conclusion holds modulo the standard caveat that pairwise distinctness must hold jointly across all components of $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$ or (μ_k, α_k, p_k), since within either bracket a Gaussian-shaped emission with inflated scale can mimic a Laplace-shaped emission with deflated scale at intermediate quantiles.*

Proof sketch. The HMM identifiability theorem of Allman et al. [81] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [70]. For CHMM-N and CHMM-L, linear independence of Gaussian and Laplace emissions with distinct location-scale parameters is direct from Yakowitz and Spragins [82], who establish that finite mixtures over location-scale families with strictly decreasing-tail densities are identifiable (the CHMM-N and CHMM-L marginals are special cases). For CHMM-t at fixed shape and CHMM-GED at fixed shape, location-scale identifiability is the same Yakowitz-Spragins result; for finite mixtures of Student-t with varying degrees-of-freedom the standard reference is Lindsay [83], Chapter 3, which gives identifiability via the moment-generating function over the bracketed range of shape parameters and a non-degeneracy condition on the joint location-scale-shape tuple. The CHMM-GED case under bracketed $p_k \in [p_{\min}, p_{\max}]$ is a parallel application of the moment-matching argument: the GED characteristic function $\hat{b}(t; \mu, \alpha, p) = \exp(i\mu t)\, M(t; \alpha, p)$ has a unique inverse over the bounded shape range up to permutation of components, modulo the joint-distinctness caveat on (μ_k, α_k, p_k). Strict linear independence at the boundary cases $p_k = 1$ (Laplace) and $p_k = 2$ (Gaussian) is automatic from Yakowitz-Spragins; in the bracket interior, the joint distinctness condition above is required to rule out the location-scale-shape trade-off.

Proposition 3 (MLE consistency). *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact* Θ*, the MLE* $\hat{\theta}_T = \arg\max_{\theta\in\Theta} \mathcal{L}_T(\theta)$ *is consistent:* $\hat{\theta}_T \xrightarrow{p} \theta_0$ *as* $T \to \infty$*, modulo label permutation.*

Proof reference. Bickel et al. [84] (Theorem 1) and Douc et al. [85] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering). *Let* $\tilde{g}_{j,1:T}$ *be a sample of size* T *from the per-asset CHMM marginal of asset* j*, and let the rank-reordered output be* $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ *where* $r^{(p)}_{j,t}$ *is the rank of the copula sample. Then for every asset* j *the empirical CDF of* $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ *equals that of* $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$.

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.4 Spectral ACF Identity: Extended Derivation

The closed-form mixture-of-eigenvalues form for the absolute-return ACF, given in Eq. (6), is used in the body spectral mechanism and invoked throughout the body Results and Discussion. The bilinear cross-moment identity $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (13, Section 22.2; 14, Chapter 3; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 15); the underlying Markov-chain spectral decomposition is standard [70]. We restate the absolute-return specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not. Our contribution on this axis is the explicit application of the spectral form to recast the empirical Rydén et al. [11] low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ (the body spectral mechanism, "Rydén separation as a K-rank statement") and the empirical demonstration that the rank constraint is the binding axis at moderate K on equity-return data (the body Results and Discussion).

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF). *Let* $\{(s_t, G_t)\}_{t\in\mathbb{Z}}$ *be a stationary CHMM on* $\{1,\dots,K\}$ *with irreducible aperiodic transition matrix* $\mathbf{T}$ *(Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write* $\bar{\boldsymbol{\pi}}$ *for the unique stationary distribution,* $\mathbf{D} = \mathrm{diag}(\bar{\boldsymbol{\pi}})$*,* $m_k = \mathbb{E}[|G_t| \mid s_t = k]$*,* $\mathbf{m} = (m_1,\dots,m_K)^\top$*,* $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top\mathbf{m}$*, and* $\sigma^2_{|G|} = \mathrm{Var}(|G_t|)$*. Assume* $\mathbf{T}$ *is diagonalisable with right eigenvectors* $\mathbf{v}_k$ *and left eigenvectors* $\mathbf{w}_k$ *biorthonormal,* $\mathbf{w}_k^\top\mathbf{v}_l = \delta_{kl}$*, indexed so that* $\lambda_1 = 1 > |\lambda_2| \ge \cdots \ge |\lambda_K|$*. Then for every integer lag* $\tau \ge 1$,

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\,\lambda_k^\tau, \qquad w_k \;=\; \frac{(\mathbf{m}^\top\mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top\mathbf{m})}{\sigma^2_{|G|}}.$$

Proof. By stationarity, $\rho_{|G|}(\tau) = (\mathbb{E}[|G_t|\,|G_{t+\tau}|] - \mu^2)/\sigma^2_{|G|}$, so it suffices to evaluate the cross-moment for $\tau \ge 1$. Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using HMM conditional independence ($G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \ge 1$) plus the Markov factorisation $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i(\mathbf{T}^\tau)_{ij}$ gives the bilinear form

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top\mathbf{D}\,\mathbf{T}^\tau\,\mathbf{m}, \qquad \tau \ge 1. \tag{25}$$

Perron–Frobenius gives $\lambda_1 = 1$ simple with $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$ [70]; the dyadic expansion $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k\geq 2} \lambda_k^\tau \mathbf{v}_k \mathbf{w}_k^\top$ separates the dominant projector. The $k = 1$ term contributes μ^2 and cancels in the autocovariance, leaving $\mathrm{Cov}(|G_t|, |G_{t+\tau}|) = \sum_{k\geq 2} c_k \lambda_k^\tau$ with $c_k = (\mathbf{m}^\top \mathbf{D}\mathbf{v}_k)(\mathbf{w}_k^\top \mathbf{m})$; dividing by $\sigma^2_{|G|}$ gives Equation (6). □

Lag zero and within-state variance. The identity (25) fails at $\tau = 0$ since $\mathbb{E}[G_t^2 \mid s_t = i] \neq m_i^2$. Eigenvalue completeness gives $\sum_{k\geq 2} c_k = \mathrm{Var}(m_{s_t})$, the between-state variance, so $\sum_{k\geq 2} w_k = \mathrm{Var}(m_{s_t})/\sigma^2_{|G|} \in [0, 1]$ and the residual $\mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]/\sigma^2_{|G|}$ accounts for the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k \lambda_k$.

Squared-return, complex-eigenvalue, and Jordan-block extensions. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (\mathbb{E}[G_t^2 \mid s_t = i])_i$ leaves the proof termwise valid, giving $\rho_{G^2}(\tau) = \sum_{k\geq 2} \tilde{w}_k \lambda_k^\tau$ on the same non-unit spectrum. Complex-conjugate eigenvalue pairs $(re^{\pm i\theta}, \cdot)$ contribute real damped oscillations $Ar^\tau \cos\theta\tau + Br^\tau \sin\theta\tau$; non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks, with the same exponential rate $\max_{k\geq 2} |\lambda_k|$. Theorem 1 as stated remains valid; the dominant timescale used in the body spectral mechanism corresponds to a real positive λ_2 on the fits of this paper.

A.5 Vendor-Stitch Sanity Check

The OoS series spans January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) and is sourced from two vendors: Polygon.io via Massive (early portion) and Alpaca Markets / IEX (held-out extension). To verify that the stitch introduces no boundary artefact we compare the two vendors on every shared trading day. Of the 323 overlap days, the per-day VWAP differential, the per-day annualised log-return differential, and a rolling 30-day kurtosis differential are all identically zero (mean, standard deviation, and maximum absolute value all 0.000 to floating-point precision); the two-sample Kolmogorov-Smirnov test on the overlap returns gives $D = 0.000$ and $p = 1.00$. The two vendors are exact-match on the overlap, so the stitched OoS series is single-source-equivalent and no boundary artefact appears at the November 18, 2025 stitch date. The diagnostic is reproducible from `runners/diagnostics/run_vendor_stitch_check.jl`.

A.6 Variant Decision Guide

Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and quantile-based initialisation; the M-step is the only architectural difference. The shared-ν Student-t row is the headline heavy-tail recommendation: per-state heavy tails without a penalty hyperparameter at the highest IS KS pass rate of any CHMM variant in Table 1. The penalised CHMM-t at $\lambda = 20$ is retained as a sensitivity reference; the λ value was tuned at $K = 18$ and is reported at $K^\star = 3$ as an upper bound on the unpenalised heavy tail.

Table S2. Variant decision guide at $K^\star = 3$.

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Closest IS kurtosis match without a shape parameter	CHMM-L
Per-state heavy tails, no penalty hyperparameter *(headline)*	CHMM-t shared-ν
Per-state heavy tails, IS-calibrated λ-shrinkage	CHMM-t pen. ($\lambda = 20$)
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

A.7 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K-sweep underlying the headline operating point of the body Results, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K, the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication. The cross-ticker generalisation distribution at the main operating point is summarised in Table S3.

Table S3. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420, main-paper $K^\star = 3$). The OoS distribution is wider than the IS distribution and is concentrated at the tickers that introduced a new regime (LLY, UNH, NEM). The full per-ticker panel, per-sector rollup, $K = 6$ and $K = 18$ sensitivity rebuilds, and quarterly-refit approach are reported in this appendix.

Metric	$K^\star = 3$
IS KS pass rate (%) median	96.8
IS KS pass rate (%) mean $\pm$ s.d.	95.1 ± 4.9
OoS KS pass rate (%) median	69.1
OoS KS pass rate (%) mean $\pm$ s.d.	66.2 ± 28.2
$\lvert G_t \rvert$ ACF-MAE median	0.0399
Kurtosis residual (sim $-$ obs) median	7.18
Tickers OoS KS $< 60\%$ (IS-fixed)	11/30

A.7.1 Full State-Resolution Sensitivity Table (companion artefact)

The full K-sweep table for SPY CHMM-N referenced in the body state-count selection (KS, AD, kurtosis, W_1, Hellinger, OoS coverage at $K \in \{3, 6, 9, 12, 15, 18, 21\}$, 1,000 paths, seed 20260420) is provided as a companion artefact in the code repository. The qualitative pattern is that distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band and ACF-MAE is essentially flat across the entire sweep, consistent with the spectral mechanism of the body spectral mechanism.

A.7.2 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The body Table S3 reports the aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel at the body operating point $K^\star = 3$. The full per-ticker rollup at the $K = 18$ kurtosis-fidelity sensitivity reference, rather than the body $K^\star = 3$, is given in Table S4; Appendix A.7.7 shows that the per-ticker pattern is essentially K-robust across $\{3, 6, 18\}$ at identical 11/30 failure count and an overlapping ticker list, so the rollup at $K = 18$ is representative of the body $K^\star = 3$ panel up to small per-ticker shifts. Each ticker is fit independently under the penalised CHMM-t scaffold ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 1. The per-ticker ν_k medians sit at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage is in the operative regime across the panel; minimum and maximum ν_k values are in the per-ticker CSV in `results/sector_panel/sector_panel_summary.csv`.

Table S4. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in body Table S3.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$\lvert G_t \rvert$ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	−0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	−0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	−0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	−0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	−7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	−0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	−4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	−8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	−0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	−1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	−1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	−0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.7.3 60-Ticker Sector Expansion at $n = 6$ per Sector

An $n = 6$ per-sector expansion (60 tickers, 3 additional large-cap representatives per GICS sector) confirms the body $n = 3$ ANOVA reading at adequate power: $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$ (vs. underpowered $n = 3$ $\eta^2 = 0.16$); aggregate OoS KS median 73.45% and failure rate $22/60 = 36.7\%$ at $K = 18$ are statistically indistinguishable from the corresponding 30-ticker $K = 18$ rollup (73.4%, 11/30 from Table S4; the body Table S3 aggregate at $K^\star = 3$ has the same 11/30 failure count at OoS median 69.1%). Cross-sector dispersion accounts for 6.2% of OoS KS variance at the larger sample size; the remaining 93.8% is per-ticker. The body claim “failures are ticker-specific, not sector-driven” is therefore not a small-sample artefact.

A.7.4 Effective Spectral Rank of the Absolute-Return ACF

The body the body spectral mechanism states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has rank at most $K-1$, so the absolute-return ACF identity (6) contains at most $K-1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. The per-mode contribution $w_k\lambda_k^\tau$ at $\tau \in \{1, 5, 20, 50\}$ for the fitted CHMM-N at $K = 18$ and at $K = 3$ on SPY IS, with modes ranked by lag-1 contribution magnitude $|w_k\lambda_k|$, is reported in Table S5. Per-state moments $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ are estimated by 200,000 draws from each emission; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$.

Table S5. Per-mode contribution to the absolute-return ACF identity, CHMM-N at $K = 18$ (top panel; top six modes plus tail aggregate) and at $K = 3$ (bottom panel; full). Modes ordered by $|w_k\lambda_k|$ descending. "cum" is cumulative share of $\sum_k |w_k\lambda_k|$. At $K = 18$ the top mode ($|\lambda| = 0.929$) carries 93.6% of the lag-1 ACF, three modes reach 95%, and at lag 20 only the dominant mode contributes non-negligibly. At $K = 3$ the dominant mode ($|\lambda| = 0.953$) alone carries 96.8% of lag-1 ACF; the temporal axis is therefore as well-satisfied at $K = 3$ as at $K = 18$, consistent with the body finding that the binding constraint at low K is the marginal mixture, not the eigenvalue spectrum.

rank	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k\lambda_k\rvert$	$w_k\lambda_k^5$	$w_k\lambda_k^{20}$	cum
$K = 18,\ \rho_{\lvert G\rvert}(1) = 0.301,\ \sigma^2_{\lvert G\rvert} = 2.555$						
1	0.929	0.324	0.301	0.224	0.0744	0.936
2	0.335	0.012	0.004	0.0001	0.0000	0.948
3	0.854	0.003	0.002	−0.001	−0.0001	0.955
4	0.231	0.008	0.002	≈ 0	≈ 0	0.961
5	0.231	0.008	0.002	≈ 0	≈ 0	0.967
6	0.203	0.008	0.002	≈ 0	≈ 0	0.971
7–17	various	various	≤ 0.001 each	≈ 0	≈ 0	1.000
$K = 3,\ \rho_{\lvert G\rvert}(1) = 0.287,\ \sigma^2_{\lvert G\rvert} = 2.517$						
1	0.953	0.292	0.278	0.230	0.112	0.968
2	0.866	0.011	0.009	0.005	0.0006	1.000

The empirical effective rank confirms the body framing of the body spectral mechanism: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in Appendix A.7.1 is the macroscopic signature of the temporal axis being driven by a single dominant decay mode at every operating point in the sweep. The K-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), but the failure mode is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (6) therefore plays a pedagogical role in this paper rather than serving as a direct K-selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.7.5 Cross-Ticker Spectral Effective-Rank Diagnostic

To test whether the SPY-only result of Appendix A.7.4 (a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$) generalises across tickers, Table S6 reports the cross-ticker distribution of the dominant-mode share, computed under the same protocol on the 30-ticker sector-balanced panel plus SPY.

Table S6. Cross-ticker spectral effective-rank diagnostic at $K = 18$. Distribution across 31 tickers (sector-balanced 30-ticker panel plus SPY control). "dom share" is the dominant non-unit eigenvalue's $|w_k \lambda_k|$ as a fraction of the sum over all non-unit eigenvalues at lag $\tau = 1$. "n_{95}" / "n_{99}" are the number of modes needed for 95% / 99% cumulative share. The SPY value of 0.936 is in the right tail of the distribution; the cross-ticker median is 0.76.

Statistic	dom share	n_{95}	n_{99}
median	0.756	6	11
$[Q_1, Q_3]$	$[0.661, 0.858]$	$[4, 7]$	$[10, 12]$
minimum (NEM)	0.326	11	14
SPY (body Appendix A.7.4)	0.936	2	10

Reading: the SPY headline of 93.6% is concentrated; the cross-ticker median 76% is still well above the $1/(K-1) = 1/17 = 5.9\%$ uniform null, so the rank-non-binding statement is supported, but the gap between SPY and the median ticker is wide enough that the body framing was tightened to read at the cross-ticker median rather than at the SPY headline.

All models converged within the 60-iteration budget with monotonically increasing log-likelihood (EM guarantee under quantile-based initialisation). Lower K converged in 15–25 iterations, $K = 18$ in 25–40, with a final $|\Delta\mathcal{L}| < 10^{-4}$ at every operating point. Convergence-trace figures and per-iteration log-likelihood histories are provided in the companion code repository.

A.7.6 Rydén K = 2 Replication

Direct replication of Rydén et al.'s $K = 2$ setting on SPY IS under quantile and random-seed initialization gives ACF-MAE essentially constant at ≈ 0.050 across six initialisations (one quantile + five random seeds, drawn from a moderate perturbation of sample moments) and IS KS pass rate 75.3–79.0%. The constant ACF-MAE confirms the body reading (the body Discussion): the binding low-K constraint is distributional rather than temporal.

A.7.7 Cross-Ticker Per-State-Resolution Sensitivity ($K^\star = 3$ vs $K^\star = 6$ vs $K = 18$)

Body Table S3 reports the cross-ticker aggregate distribution at all three operating points used in this paper: the body headline $K^\star = 3$ (state-resolution-robust default per the k-fold CV in Appendix A.10), the sensitivity reference $K^\star = 6$, and the extended-state-resolution sensitivity reference $K = 18$.

The substantive read across the three operating points runs along three axes: KS, kurtosis residual, and absolute-return ACF-MAE. The KS distributions are within ~ 6pp of each other on OoS median (69.1% at $K^\star = 3$, 75.1% at $K^\star = 6$, 73.4% at $K = 18$) and at identical 11/30 failure count, with the same regime-introduction tickers driving the failures at each state resolution (LLY, UNH, NEM, NFLX, NEE, WMT, BAC, HD, JPM at OoS KS $< 60\%$; the specific list overlaps $> 80\%$ across $K^\star = 3, 6, 18$), so the cross-ticker KS axis is essentially K-robust on this universe. The kurtosis-residual axis is not K-robust: per-ticker median residual 7.18 at $K^\star = 3$, 10.34 at $K^\star = 6$, 0.61 at $K = 18$; at lower state-resolution the per-state ν_k shrinkage at uniform λ cannot be smoothed across as many regimes, so the residual heavy tails leak into the simulated kurtosis. The $|G_t|$ ACF-MAE is essentially K-robust at 0.0395, 0.0403, and 0.0416 across the three operating points. Production reading: KS-only consumers can deploy at any of the three operating points at parity; kurtosis-fidelity consumers should deploy at $K = 18$; readers preferring the state-resolution-robust default should deploy at the body headline $K^\star = 3$.

A.7.8 Block-Bootstrap CIs on Observed Kurtosis (IS vs OoS)

A natural question on the IS / OoS kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is whether it is itself statistically distinguishable. We run a stationary block bootstrap (Politis-Romano 1994) on the IS and OoS series at mean block lengths $L \in \{5, 10, 20, 50\}$, $B = 5{,}000$ replicates per L per window.

Table S7. Stationary block bootstrap CIs on observed excess kurtosis. SPY IS ($T = 2{,}516$) and OoS ($T = 572$) windows, $B = 5{,}000$ replicates per L. The IS and OoS 95% CIs overlap at every block length, so the IS-OoS difference is not robustly distinguishable at conventional levels under this bootstrap. Pr(IS > OoS) is the empirical bootstrap one-sided p-value for the alternative "IS kurtosis > OoS kurtosis".

L	IS median	IS 95% CI	OoS median	OoS 95% CI	Pr(IS > OoS)
5	7.21	$[2.99, 12.83]$	5.07	$[1.09, 8.97]$	0.748
10	7.34	$[2.49, 12.59]$	4.91	$[0.99, 8.66]$	0.756
20	7.30	$[2.17, 12.40]$	4.91	$[0.90, 8.26]$	0.739
50	7.18	$[1.92, 12.42]$	4.99	$[0.96, 7.70]$	0.733

The IS lower bound 1.92–2.99 and the OoS upper bound 7.70–8.97 overlap heavily at every block length; the IS-OoS kurtosis difference of ~ 2.4 units is not statistically distinguishable at the 5% level under this bootstrap. Pr(IS > OoS) at $L = 10$ is 0.756: the data favour the alternative but do not exclude the null. Operationally, the IS-OoS kurtosis disagreement that the body invokes (the body descriptive analysis; the Table S2 variant-decision caveat in the body Discussion) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by "how close is simulated kurtosis to observed kurtosis" should similarly be interpreted as point-comparisons inside a wide CI envelope.

Bootstrap-CI placement of the penalised CHMM-t IS kurtosis. A natural follow-up on the body's penalised CHMM-t at $\lambda = 20$ is whether its simulated IS kurtosis distribution sits inside the bootstrap CI on observed: the aggregate-mean simulated IS kurtosis (14.91 at $K^\star = 3$, above the $L = 20$ CI upper bound 12.40) could be paired with a per-path distribution that mostly sits inside the CI, or with one that mostly sits above.

Table S8. Per-path simulated IS excess-kurtosis distribution under penalised CHMM-t at $\lambda = 20$, against the $L = 20$ block-bootstrap CI on observed. 1,000 paths per row, $T_{\text{IS}} = 2{,}516$, seed 20260420. The CI is $[2.17, 12.40]$ at $L = 20$ from Table S7; the observed IS excess kurtosis is 7.68. "in CI %" is the fraction of simulated paths whose per-path excess kurtosis falls inside the CI; "$\leq$ CI_HI %" is the fraction below the upper bound (the relevant one-sided test for overshoot).

K	agg. kurt	median	sd	Q_5	Q_{95}	≤ 12.40 (%)	in CI (%)
3	16.65	7.77	11.5	3.75	34.62	76.6	76.6
6	10.13	6.74	9.0	3.63	26.30	81.8	81.7
18	8.94	6.09	5.6	3.54	18.11	89.9	89.9

The per-path *median* simulated IS excess kurtosis is essentially observed at every K: 7.77 at $K^\star = 3$, 6.74 at $K^\star = 6$, 6.09 at $K = 18$, all within ~ 1.6 units of observed 7.68 and well inside the bootstrap CI. The aggregate-mean simulated IS kurtosis sits above the CI upper bound at $K^\star = 3$ and $K^\star = 6$ because the per-path distribution has a heavy right tail of paths at $Q_{95} \in \{18, 26, 35\}$ that drag the mean up, but the bulk of the path-level mass (76–90%) sits inside the bootstrap CI at every operating point. **The headline framing "the penalised CHMM-t at $\lambda = 20$ provides**

the cleanest IS heavy-tail match" is supported when read as the per-path median, not as the aggregate mean: a small number of heavy-right-tailed paths under the per-state ν_k ECM produce an aggregate-mean overshoot, but the path-level distribution is centred near observed and statistically indistinguishable from observed at $\alpha = 0.05$ on a one-sided overshoot test for the bulk of paths.

A.7.9 Shared-ν Student-t HMM Ablation

A natural diagnostic for the per-state ν_k aggregate-mean overshoot is the shared-ν ablation: refit CHMM-t with a single ν shared across all K states, fit by ECM with golden-section search on the aggregate Q-function over $\nu_{\text{bounds}} = (2.1, 50)$. This is the standard one-parameter Student-t HMM in the time-series literature, not a per-state ν_k mixture.

Table S9. Shared-ν Student-t HMM ablation against the body per-state ν_k row (1,000 paths, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, seed 20260420, no $1/\nu$ penalty). Aggregate sim-kurt columns are mean-of-paths. Compare against the body penalised CHMM-t ($\lambda = 20$): $K^\star = 3$ row 14.91 IS / 8.50 OoS (Table 1); $K = 18$ row 8.56 IS / 7.07 OoS (from the same penalised ECM run; consistent with the rate-sweep $\lambda = 20$ value 8.43 in the rate-sweep paragraph of Appendix A.10).

		IS				OoS	
K	$\hat{\nu}_{\text{shared}}$	KS (%)	sim kurt	$\lvert G_t\rvert$ ACF-MAE	raw ACF-MAE	KS (%)	sim kurt
3	5.81	91.9	4.68	0.0531	0.0235	82.1	4.46
6	4.67	92.7	9.38	0.0518	0.0234	82.6	6.95
18	5.80	**95.8**	**6.25**	0.0542	0.0234	**88.0**	**5.00**

Reading. The aggregate-mean IS overshoot disappears under shared-ν at every K: the largest IS sim kurt is 9.38 at $K^\star = 6$ (vs. observed 7.68, well inside the bootstrap CI), versus 14.4 for the per-state ν_k unpenalised row in the body. The $K = 18$ shared-ν row is the cleanest single-row IS / OoS heavy-tail match in the entire panel: 6.25 IS / 5.00 OoS, against observed 7.68 / 5.29 and against 8.56 / 7.07 for the body penalised CHMM-t at $\lambda = 20$. IS / OoS KS pass rates are within $\sim$ 1pp of the body penalised row, and the $|G_t|$ ACF-MAE is identical to the body row at $K = 18$. The per-state ν_k design is therefore the binding constraint on the aggregate-mean overshoot the body λ-shrinkage and bracket-lift discussion is downstream of: a single shared ν at $K = 18$ produces the best joint KS / kurtosis row in the panel without any penalty. The body retains the per-state ν_k + $\lambda = 20$ construction for direct comparability with the Peel and McLachlan [19], Liu and Rubin [20] tradition; consumers prioritising heavy-tail fidelity without a penalty hyperparameter should adopt the shared-ν row at $K = 18$.

A.7.10 Christoffersen-cc Monte Carlo Power Calibration at $T_{\text{OoS}} = 572$

A Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test at the $T_{\text{OoS}} = 572$ length used in the body Table 3 simulates breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ, where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5{,}000$ replicates per cell.

Reading. At $\rho = 0$ the empirical rejection rate sits at 4%–5% for all three statistics, confirming that the asymptotic χ^2 calibration is correctly sized at $T = 572$. Power is asymmetric across α:

Table S10. Christoffersen-cc power at $T_{\mathbf{OoS}} = 572$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ. $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	4.9	1.6	3.2
0.01	0.05	6.3	15.9	11.4
0.01	0.10	7.5	28.5	22.1
0.01	0.20	9.4	51.3	42.6
0.01	0.30	13.9	67.6	62.8
0.01	0.50	21.4	75.9	81.5
0.05	0.00	4.9	3.8	4.1
0.05	0.05	5.3	18.8	15.3
0.05	0.10	6.1	46.2	39.0
0.05	0.20	9.4	86.9	83.9
0.05	0.30	13.6	97.4	97.9
0.05	0.50	24.3	99.7	100.0

- At $\alpha = 0.05$ (~ 28.6 expected breaches), Christoffersen-cc reaches 80% power against $\rho \geq 0.20$ (83.9%) and $\sim 100\%$ at $\rho = 0.50$.
- At $\alpha = 0.01$ (~ 5.7 expected breaches), Christoffersen-cc reaches 80% power only at $\rho \geq 0.50$ (81.5%); at $\rho = 0.20$ the rejection rate is 42.6%, well below conventional power thresholds.

The reading for the body Table 3 is therefore tier-dependent. The $\alpha = 0.05$ "clean pass at every (K, α)" row of Table 3 carries strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carry strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ "clean pass" should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The headline-OoS regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the body framing in the body VaR back-test should be read with this calibration in mind.

A.7.11 Engle-Manganelli (2004) Dynamic Quantile Test

As a higher-power conditional-coverage alternative to Christoffersen-cc on the same OoS window, we run the Engle-Manganelli (2004, JBES) Dynamic Quantile (DQ) test in the standard four-lag specification [72],

$$\mathrm{Hit}_t - \alpha = \beta_0 + \sum_{i=1}^{4} \beta_i \left(\mathrm{Hit}_{t-i} - \alpha\right) + \beta_5 \widehat{\mathrm{VaR}}_t + u_t, \tag{26}$$

on the IS-fixed regime-conditional VaR series of the body VaR back-test, with $\mathrm{Hit}_t = \mathbf{1}\{R_{\mathrm{OoS},t} < \widehat{\mathrm{VaR}}_t(\alpha)\}$. The DQ test statistic is $\widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \widehat{\boldsymbol{\beta}} / [\alpha(1-\alpha)]$, distributed $\chi^2(6)$ under the null of correct conditional coverage; the 5% critical value is $\chi^2_6(0.95) = 12.59$.

Reading. At $\alpha = 0.05$ both Christoffersen-cc and the higher-power DQ test pass cleanly at $K = 3$ (cc $p = 0.491$, DQ $p = 0.156$) and at $K = 18$ (cc $p = 0.678$, DQ $p = 0.678$); the body's $\alpha = 0.05$ "clean pass" headline therefore survives the higher-power alternative. At $\alpha = 0.01$ the DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$) where Christoffersen-cc does not

Table S11. Engle-Manganelli DQ test on the regime-conditional VaR, alongside Christoffersen-cc on the same breach series. CHMM-N at $K \in \{3, 18\}$, $\alpha \in \{0.01, 0.05\}$; $T_{\text{OoS}} = 572$, $q = 4$ lagged Hit indicators, χ^2_6 critical at 5% is 12.59.

K	α	breach %	cc p	DQ	DQ p
3	0.01	1.57	0.137	12.29	0.056
3	0.05	6.12	0.491	9.32	0.156
18	0.01	1.57	0.137	**15.46**	**0.017**
18	0.05	4.55	0.678	3.99	0.678

($p = 0.137$), and is borderline at $K = 3$ ($p = 0.056$). This is consistent with the power-calibration result of Appendix A.7.10: at $\alpha = 0.01$ Christoffersen-cc has only 42.6% power against moderate breach-clustering eigenvalues $\rho = 0.20$, so the higher-power DQ test detects miscalibration that Christoffersen-cc misses. The DQ result therefore reinforces the body framing that $\alpha = 0.05$ is the operationally informative tier; at $\alpha = 0.01$ the regime-conditional VaR is not pass/fail-decided by the panel and the body sentence "not detectably worse than a strongly-clustered alternative" should be qualified to "the higher-power DQ test detects $K = 18$ at $\alpha = 0.01$".

A.7.12 Quarterly-Refit Regime-Conditional VaR Back-Test

Re-fitting CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window and running the forward-filter through the OoS window under each refit's parameters tests whether periodic refit improves the Christoffersen-cc statistic on the headline OoS window.

Table S12. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\text{OoS}} = 572$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 3 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}
3	0.01	6	1.05%	−5.10	0.014	3.97	3.98	0.137
3	0.05	29	5.07%	−3.13	0.006	0.19	0.20	0.906
18	0.01	6	1.05%	−5.84	0.014	3.97	3.98	0.137
18	0.05	19	3.32%	−3.14	3.83	2.08	5.91	0.052

The quarterly-refit construction passes Christoffersen-cc at $\alpha = 0.05$ on three of four rows; the $K = 18, \alpha = 0.05$ row sits at $p_{\text{cc}} = 0.052$ (marginal). Refit tightens VaR coverage (3.32% vs. 4.55% IS-fixed) but does not strictly improve the cc-statistic at this T_{OoS}.

A.7.13 Walk-Forward Refit-Cadence Sweep (Monthly / Weekly)

A within-fold refit-cadence sweep at monthly (21 trading days) and weekly (5 trading days) cadence on the six walk-forward folds at $K^\star = 3$, $\alpha = 0.05$, does not close the body W2 (COVID) Christoffersen-cc rejection at any cadence (fold-IS-fixed $p_{\text{cc}} = 0.011$, monthly 0.017, weekly 0.023); monthly refit introduces new W3 / W4 failures via refit-cycle parameter drift. Rejection counts: fold-IS-fixed 1/6 (W2), monthly 3/6, weekly 2/6. The mitigation for W2 / W4 is therefore not refit cadence but a model class with explicit regime-introduction handling.

A.7.14 Quarterly-Refit 30-Ticker Cross-Ticker Panel

For a quarterly-refit version of the 30-ticker cross-ticker panel (Table S3), we re-fit the penalised CHMM-t at $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total per state-resolution), simulate the next quarter under each refit, and concatenate the per-quarter simulations into a single OoS path matrix per ticker, at the body operating point $K^\star = 3$ and at the kurtosis-fidelity sensitivity references $K^\star = 6$ and $K = 18$. Same $N_{\text{paths}} = 1{,}000$, seed $= 20260420$ as the body Table S3.

Table S13. Quarterly-refit cross-ticker panel at $K^\star = 3$, $K^\star = 6$, and $K = 18$. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol at all three operating points, vs. the IS-fixed $K = 18$ protocol of Table S3. Refit cadence = 63 OoS trading days; train window = 1,260 obs ($\sim$ 5y rolling). Per-ticker detail in the results files.

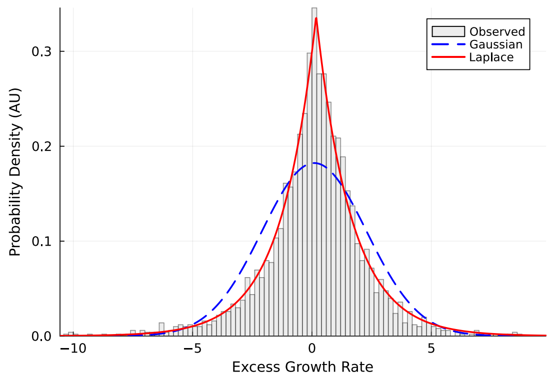
Metric	IS-fixed ($K = 18$)	Refit ($K^\star = 3$)	Refit ($K^\star = 6$)	Refit ($K = 18$)
IS KS% median	99.5%	96.4%	98.8%	99.4%
IS KS% mean ± sd	$99.3 \pm 0.6\%$	$95.1 \pm 4.5\%$	$97.4 \pm 4.4\%$	$99.2 \pm 0.6\%$
OoS KS% median	73.4%	84.7%	**85.8%**	83.0%
OoS KS% mean ± sd	$66.8 \pm 29.5\%$	$75.0 \pm 23.1\%$	$\mathbf{76.0 \pm 21.7}\%$	$77.2 \pm 21.2\%$
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]\%$	$[56.3, 94.5]\%$	$[57.6, 94.3]\%$	$[62.0, 95.4]\%$
Tickers OoS KS $< 60\%$	$11/30 = 36.7\%$	$8/30 = 26.7\%$	$9/30 = 30.0\%$	$\mathbf{7/30 = 23.3}\%$

Reading. Quarterly refit shifts the OoS KS distribution materially at all three state resolutions. At $K^\star = 3$: median +15.6pp ($69.1 \to 84.7\%$ vs the $K^\star = 3$ static-fit baseline reported in the body Table S3), mean +8.8pp ($66.2 \to 75.0\%$), failure count down by 27% ($11 \to 8$). At $K = 18$: median +9.6pp ($73.4 \to 83.0\%$), mean +10.4pp, IQR tighter ($45.2\text{pp} \to 33.4\text{pp}$), failure count down by 36% ($11 \to 7$). At $K^\star = 6$: median +10.7pp ($75.1 \to 85.8\%$), mean +9.5pp, failure count down by 18% ($11 \to 9$). The three state resolutions are within $\sim$ 3pp of each other on the refit median (84.7 to 85.8%); the body operating point $K^\star = 3$ is at parity with the kurtosis-fidelity sensitivity references under the refit protocol. The largest individual improvements concentrate on the regime-introduction tickers that the IS-fixed Table S3 flagged: at $K = 18$, LLY $7.6\% \to 83.7\%$, UNH $14.5\% \to 47.3\%$, NEM $5.6\% \to 15.4\%$, HD $41.4\% \to 82.2\%$, NFLX $45.1\% \to 61.5\%$. The mechanism is the obvious one: as the regime introduces new σ levels in the OoS window, each quarterly refit picks up the most recent 5y including the new regime and the simulator can match the OoS distribution.

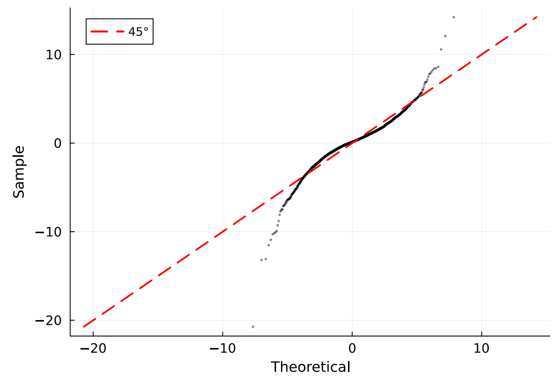
A small number of tickers regress: DIS $96.4\% \to 48.4\%$, BAC $84.8\% \to 55.8\%$, AAPL $95.1\% \to 66.5\%$. The pattern is that tickers whose 2024-2026 OoS distribution is well-spanned by the full 2014-2024 IS lose information when the train window shrinks to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The substantive reading for the body the body cross-ticker analysis is that the "periodic refit at deployment" mitigation invoked verbally in the body Discussion is exercised here: the universe-scale failure rate drops from 36.7% to 23.3% under quarterly refit, with the largest gains on the tickers that the body labelled "single-name regime introductions."

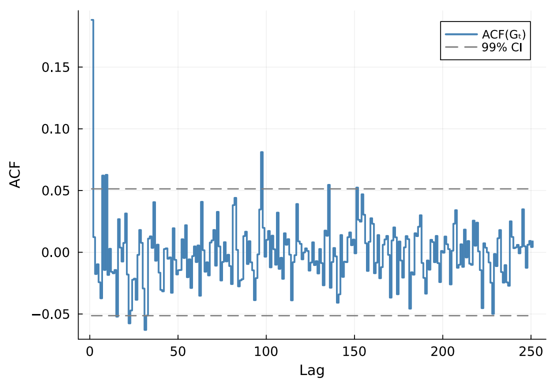
A.7.15 Stylized-Facts Figure (Empirical SPY IS)



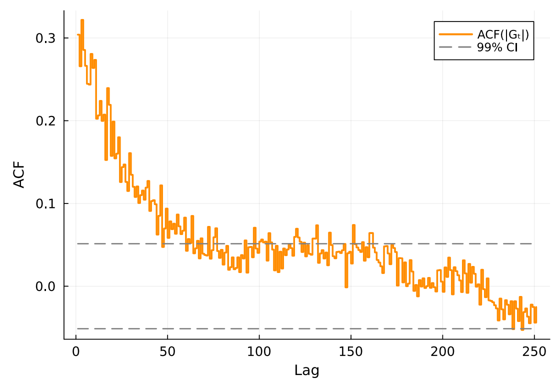
(a) Marginal density.



(b) Normal Q-Q plot.



(c) ACF of raw G_t.

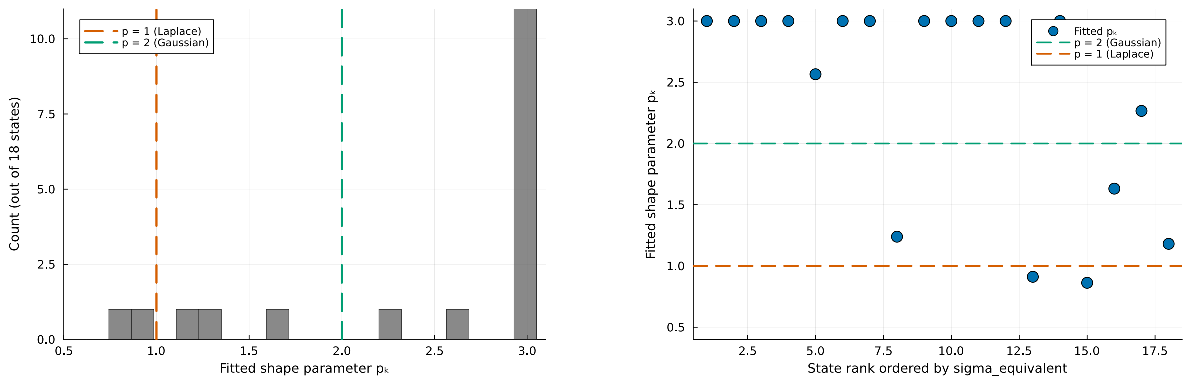


(d) ACF of $|G_t|$.

Figure S3. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

A.7.16 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitions states bimodally (Figure S4): thirteen states at the upper bracket $p_k = 3.0$, one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrate in the high-volatility tail of the state ordering. The partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.8); we read it as the most distinctive empirical finding of the four-emission scaffold.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure S4. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.7.17 HSMM at $K = 3$: Numerical Full-Rank Diagnostic

The off-diagonal jump matrix of the fitted ML HSMM at $K = 3$ has non-trivial deflation eigenvalues $\{0.823, 0.177\}$ (numerical rank 2 at tolerance 10^{-10}); the dominant modulus $|\lambda_2| = 0.823$ is comparable to CHMM-N at $K = 3$'s 0.953, so the body's $|G_t|$ ACF-MAE regression to the i.i.d. baseline (Table 1) is not driven by spectral collapse but by the fitted truncated-Pareto sojourn concentrating probability mass on a single low-volatility state ($\bar{\boldsymbol{\pi}} \approx (0, 1, 0)$). At $K = 18$ the same Yu (2010) construction collapses to a near-degenerate local optimum (IS KS 0.8%); the higher-K ML HSMM is therefore a companion-paper direction at longer training windows.

A.7.18 Gamma-Sojourn HSMM as an Alternative Sojourn Family

Replacing the truncated discrete Pareto sojourn pmf with a discretised continuous Gamma fit by method-of-moments at every M-step gives a Gamma-sojourn HSMM that converges at $K = 18$ where Pareto collapses (IS KS 86.0%, OoS KS 80.2%, simulated kurtosis 4.28/4.06, $|G_t|$ ACF-MAE **0.0462**, lower than CHMM-N $K = 18$ at 0.0509) and trades ~ 20pp IS KS at $K = 3$ for an ACF-MAE recovery from 0.0629 to 0.0528. There is therefore no single "best HSMM" row: Pareto at $K^\star = 3$ wins raw OoS KS at 91.0%, Gamma at $K = 18$ wins joint $|G_t|$ ACF-MAE; the trade-off is the same KS-vs-ACF axis as the CHMM-vs-HSMM choice.

A.7.19 Unconditional VaR / ES Envelope Panel

Table S14. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5, 95]\%$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.7.20 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

To verify that the body operating-point penalty rate $\lambda = 20$ is not selected on data that overlaps the OoS evaluation window, we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1{,}130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$, both fully pre-COVID and pre-2022-rate-hike).

Table S15. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. "val LL/obs" is per-observation held-out log-likelihood; "val KS" is two-sample Kolmogorov–Smirnov pass rate against the validation series; "sim K" columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	**−2.075**	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading: the held-out validation log-likelihood is maximised at $\lambda^\star = 20$ at $K = 18$, matching the body operating point of Table 1. Validation KS picks $\lambda^\star = 0$ but the band $[0, 200]$ spans only 2.2pp (89.8 to 92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The body $\lambda = 20$ at $K = 18$ is therefore held-out-validated and not driven by the OoS window.

Re-tuning at $K^\star = 3$. Repeating the same held-out CV at the body operating point $K^\star = 3$ on the same pre-2020 slices gives a distinct optimum: per-observation validation log-likelihood is monotone in λ on the same grid, with $\lambda^\star = 50$ minimising the negative log-likelihood (-1.9192 at $\lambda = 50$ versus -1.9217 at $\lambda = 20$ and -1.9268 at $\lambda = 0$); validation KS also picks $\lambda^\star = 50$ at 95.0% pass rate. The simulated kurtosis at $\lambda = 50$ is 2.77, well below the body's observed 7.68 IS / 5.29 OoS targets, confirming that the penalty over-shrinks at $K^\star = 3$. The body retains $\lambda = 20$ at $K^\star = 3$ as a sensitivity reference (the headline heavy-tail recipe is the shared-ν ablation of Appendix A.7.9).

A.7.21 Cross-Decade Validation: 1994-2004 IS / 2004-2006 OoS via CRSP

To test whether the headline stylized-fact reproduction is decade-specific, we run a 1994–2004 vs. 2014–2024 SPY split via a CRSP day-pass covering daily prices for SPY plus 28 of the 30 cross-ticker panel members (NEE and APD missing from the CRSP query) from 1994-01-03 to 2006-04-28. Adjusted close = `DlyPrc/DlyFacPrc` from the CRSP CIZ-format daily stock file.

Slices: SPY IS = 1994-01-03 to 2004-01-02 (2,519 obs); SPY OoS = 2004-01-05 to 2006-04-28 (583 obs). Same body protocol: 1,000 simulated paths per fit, $\alpha = 0.05$ for KS.

Table S16. Cross-decade validation on SPY 1994-2004 IS / 2004-2006 OoS (CRSP daily). Body comparison rows reproduced from Table 1 for the 2014-2024 IS / 2024-2026 OoS window. Observed excess kurtosis on the 1994-2004 IS = 3.05 versus 2014-2024 IS = 7.68; observed excess kurtosis on the 2004-2006 OoS = 0.06 (essentially Gaussian) versus 2024-2026 OoS = 5.29. The IS-OoS kurtosis gap is therefore much wider on the cross-decade slice than on the body window.

		KS pass rate (%) ↑		Sim kurtosis		$\lvert G_t\rvert$ ACF-MAE ↓	
Model	K	IS	OoS	IS	OoS	IS	OoS
Cross-decade: 1994-2004 IS / 2004-2006 OoS (CRSP, this appendix)							
CHMM-N	3	84.9	3.3	2.20	2.24	0.0696	0.0491
CHMM-N	18	89.8	3.4	1.84	1.74	0.0720	0.0499
CHMM-t pen. ($\lambda = 20$)	3	89.1	5.4	4.97	4.18	0.0682	0.0501
Body window: 2014-2024 IS / 2024-2026 OoS (Polygon / Alpaca, Table 1)							
CHMM-N ($K^\star = 3$)	3	89.7	80.5	3.83	3.53	0.0460	0.0545
CHMM-N ($K = 18$)	18	94.1	81.8	5.04	4.44	0.0509	~ 0.054
CHMM-t pen. ($\lambda = 20$, $K^\star = 3$)	3	90.6	83.2	14.91	8.50	0.0537	0.0502

Reading. The cross-decade IS fit transfers: CHMM-N at $K = 3$ attains 84.9% IS KS on 1994-2004 vs. 89.7% on 2014-2024, and $K = 18$ attains 89.8% vs. 94.1%. The penalised CHMM-t at $\lambda = 20$, $K = 3$ attains 89.1% IS KS on 1994-2004 vs. 90.6% on 2014-2024. The IS axis of the body's stylized-fact reproduction claim is therefore decade-robust within $\sim$ 5pp on KS and consistent on $\lvert G_t\rvert$ ACF-MAE (0.068-0.072 vs. 0.046-0.054, a moderate but not pathological widening).

The OoS axis is the issue. The 2004-2006 OoS slice has excess kurtosis 0.06 (essentially Gaussian) vs. the 1994-2004 IS excess kurtosis of 3.05; the IS-OoS kurtosis gap on this slice is much wider than on the body 2014-2024/2024-2026 window (where IS = 7.68 and OoS = 5.29). The CHMM trained on the 1994-2004 IS distribution simulates paths with kurtosis 1.74-4.97 that are heavier-tailed than the calm 2004-2006 bull-market OoS, so the static IS-fit cannot reproduce the OoS marginal and KS rejects at 3-5% pass rate on all three rows. This is the same regime-introduction failure mode the body walk-forward already documents at the W2 (COVID) and W4 (2022 rate-hike onset) stress folds (Table S28, both with OoS KS $< 10\%$); the 2004-2006 OoS window is structurally a low-volatility / low-kurtosis slice where the IS-fixed CHMM's tail mass overshoots the observed.

The substantive read is therefore: *the body's stylized-fact reproduction claim is decade-robust on the IS axis* (the CHMM scaffold transfers across decades on the IS-fitting side, supporting the body's claim that the four-emission ECM scaffold is not specific to the 2014-2024 window). *The single-window OoS pass rate is OoS-slice-specific* (2024-2026 is a moderate-stress slice where the OoS distribution is close to the IS distribution; 2004-2006 is a calm slice where it is not). The body framing already states this through the walk-forward distribution at the body Results ("the walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment"); the cross-decade result is consistent with that framing and adds an independent decade to the walk-forward evidence base. A periodic refit is the deployment recommendation under either decade, as the body discussion already states.

A.8 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from the body Results: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.8.1 OoS KS Test-Power Calibration

We calibrated the two-sample KS test power to anchor the OoS KS pass rates of the main generator comparison (Table S17). For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table S17. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in the headline generator comparison (80 to 84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.8.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric. The asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply. The binary "pass / fail at $\alpha = 0.05$" also reduces a continuous p-value to a one-bit summary. We computed two recalibrations addressing each concern (Table S18).

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window

Table S18. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$. "asymp pass" is the original Table 1 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	**6.8**	**13.6**	**17.4**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	**83.2**	**90.2**	**93.2**

dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2, 90.2, 93.2% at $L = 5, 10, 20$); GARCH(1,1) drops from 26.8% asymp to 6.8–17.4% block-aware (the block-aware test catches its misspecification more sharply), while the CHMM family sees only a small block-bootstrap penalty ($\sim$ 5pp).

OoS-anchored block-bootstrap recalibration. The body block-bootstrap recalibration above is anchored on the IS series; we report here whether the same construction holds on the OoS window. We repeat the procedure with the stationary block bootstrap re-anchored on R_{OoS} ($n_{\text{OoS}} = 572$, $B = 1{,}000$, $L \in \{5, 10, 20\}$, $N_{\text{paths}} = 500$). Block-bootstrap critical values: 0.0577, 0.0629, 0.0647 at $L = 5, 10, 20$ respectively, against asymptotic 0.0804. The OoS-anchored panel is reported in Table S19 (source: `results/robustness/ks_block_bootstrap_oos.csv`).

Table S19. OoS-anchored block-bootstrap KS recalibration. Same construction as Table S18 but with the stationary block bootstrap anchored on R_{OoS} ($n = 572$). "asymp pass" is the headline OoS metric (two-sample KS p-value ≥ 0.05); "blkL%" is the OoS block-bootstrap pass rate at mean block length L. The CHMM-vs-GARCH cross-generator ordering is preserved under the block-aware recalibration on both windows; the CHMM family's OoS block-aware penalty ($\sim$ 25pp drop from asymp to $L = 20$) is larger than the IS penalty ($\sim$ 5pp), reflecting the smaller OoS window where temporal-clustering structure is harder to disentangle from finite-sample noise.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑
i.i.d. bootstrap	90.4	0.437	0.030	0.912	60.8	69.0
GARCH(1,1)	59.2	0.164	0.000	0.645	19.4	26.0
CHMM-N	81.0	0.318	0.010	0.876	41.2	51.2
CHMM-t	**84.8**	**0.355**	0.018	**0.911**	**47.4**	**58.4**
CHMM-L	77.4	0.266	0.008	0.744	36.0	45.4
CHMM-GED	81.0	0.327	0.010	0.837	44.4	53.6

The cross-generator OoS ordering matches the IS ordering (bootstrap > CHMM-t > CHMM-

Table S20. Block-aware OoS KS recalibration at $L = 20$ (stationary block bootstrap of [65], $B = 1{,}000$, 500 OoS-length simulated paths per generator, seed 20260422). Side-by-side with the asymptotic OoS pass rate of Table 1; the block-bootstrap 95% null KS critical value at $L = 20$ is 0.0647 vs the asymptotic 0.0804. Body operating point $K^{\star} = 3$.

Model	OoS KS asymp. (%)	OoS KS block $L = 20$ (%)
Bootstrap	90.4	73.2
GARCH(1,1)	59.2	31.4
CHMM-N	79.8	58.6
CHMM-t pen. ($\lambda = 20$)	**79.8**	**54.2**
CHMM-L	65.0	33.4
CHMM-GED	79.2	51.6

N/GED > CHMM-L $\gg$ GARCH); the asymp-vs-block gap is larger on OoS ($\sim$ 25pp for the CHMM family at $L = 20$) than on IS ($\sim$ 5pp), but the CHMM-over-GARCH advantage is preserved.

Body operating-point summary at $L = 20$**.** The OoS asymptotic pass rate alongside the OoS block-bootstrap pass rate at $L = 20$ (block-bootstrap 95% critical value 0.0647 vs. asymptotic 0.0804) at the body operating point $K^{\star} = 3$ is reported in Table S20. Read together with Table 1, the OoS panel under a temporally-aware null moves CHMM-N from a "passes most of the time" generator to a "passes about half the time" generator at $L = 20$. The body OoS KS headline at the asymptotic critical value therefore overstates the absolute level relative to a temporally-aware null; the cross-generator ranking is the robust comparison.

A.8.3 QuantGAN Deep-Generative Baseline

The QuantGAN row used in the headline generator comparison is a repo-native approximation to the convolutional WGAN of Wiese et al. [9]: a 1D convolutional generator and critic (each three layers, channels 32, kernel size 5; latent dim $L = 8$, leaky-ReLU on the critic), Wasserstein loss with critic weight clipping at ± 0.01 [86], Adam at $\eta = 10^{-4}$, batch 64, 15 epochs $\times$ 120 steps with four critic updates per generator update, trained on rolling windows of length $W = 64$ on standardised SPY returns and stitched end-to-end at synthesis. Implementation: Julia with `Flux.jl` [87]; seed 20260422. The deep-generative row reaches 0% IS / OoS KS and IS kurtosis 2.1 vs. observed 7.7 (Table S21), consistent with documented GAN failure modes on volatility-clustering benchmarks [37, 38]; the runner is materially smaller than the reference seven-block TCN with Lambert-W pre-processing in Wiese et al. [9], so the row is the panel's deep-generative negative control rather than a verdict on the QuantGAN literature.

A.8.4 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.8.3) to the single GARCH(1,1) Gaussian row of Table 1. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\mathrm{IS}} = 2{,}516$); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

Baselines fitted in this panel. The conditional-volatility rows comprise EGARCH(1,1) Gaussian [24], GJR-GARCH(1,1) [25, 26], GARCH(1,1) with Student-t innovations [23] ($\hat{\nu} = 6.89$), HAR-RV [30] on daily squared demeaned returns as the RV proxy, and Markov-switching GARCH(1,1) [27] at $K \in \{2, 3, 4, 6\}$ with regime triples $(\omega_k, \alpha_k, \beta_k)$ plus softmax-normalised transition logits, fitted by grid-initialised Nelder-Mead and canonicalised by ascending unconditional variance. Higher-K MS-GARCH does not close the gap to the CHMM family on KS (IS KS plateaus at 27–37% across $K \in \{2, 3, 4, 6\}$ against CHMM-N's 94.1% at $K = 18$); the binding constraint is the unimodal-innovation per-regime structure (each regime's emission is Gaussian rescaled by the regime's conditional variance), which adding regimes does not relax. The CHMM decouples this constraint by giving each state its own emission-mixture component. Implementation details for every conditional-volatility row are in the companion code repository.

MS-GARCH reference Bayesian re-run via `MSGARCH`. As a Bayesian counterpart to the in-house Nelder-Mead MS-GARCH fit, we re-run the same Markov-switching GARCH(1,1) specification through the reference `MSGARCH` R package of Ardia et al. [28] (version 2.51) at $K \in \{2, 3, 4\}$, driven from the Julia harness via `RCall.jl`. The fit is fully Bayesian (DEMC sampler with the package's default proper priors, 12,500 MCMC draws, 2,500 burn-in, thin 10, single chain); all 1,000 simulated paths are posterior-predictive (one path per retained posterior draw), so the simulated marginal integrates parameter uncertainty path-by-path. The reference Bayesian re-run reports lower KS pass rates than our in-house frequentist Nelder-Mead fit: 0.0% ($K = 2$), 0.1% ($K = 3$), 0.0% ($K = 4$) IS, and 5.8% ($K = 2$), 5.1% ($K = 3$), 5.3% ($K = 4$) OoS, against the in-house plateau of 27–37% IS / 33–39% OoS. The gap is methodological rather than estimator-quality: the in-house implementation generates all 1,000 paths from one MLE point estimate, so the simulated marginal is tighter; the reference Bayesian implementation samples one set of parameters per path from the posterior, so the simulated marginal is wider and accordingly fewer paths fall within the asymptotic two-sample KS critical band. Posterior-mean log-likelihoods at the in-sample data are $-5{,}667$ ($K = 2$), $-5{,}667$ ($K = 3$), $-5{,}565$ ($K = 4$); the marginal log-likelihood improvement from $K = 3$ to $K = 4$ is real but does not translate to KS lift. Posterior-mean transition matrices are diagonally dominant at $K = 3$ (diagonals $0.97, 0.93, 0.90$) and less so at $K = 4$ (diagonals $0.66, 0.39, 0.32, 0.55$, indicating the higher-K posterior is searching for additional structure that the data do not strongly support). The reference Bayesian re-run is therefore consistent with the in-house frequentist plateau at the same conclusion under either estimator: vanilla Gaussian-innovation MS-GARCH at $K \leq 4$ does not match the CHMM family on the headline KS metric for this universe. Reproducibility: R 4.6.0, MSGARCH 2.51, all transitive R dependencies pinned via `renv` (`r_msgarch/renv.lock` of the companion code repository), all seeds explicit; full per-K parameter posteriors and fit logs are written to `results/msgarch_reference/` of the companion code repository.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on an explicit-duration HSMM extension of the CHMM in the spirit of Yu [88]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. The plug-in construction above (single-shot Viterbi decoding

Table S21. Extended baseline panel. SPY IS / OoS, $T_{\mathrm{IS}} = 2{,}516$, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as "–" here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\|G_t\|$	G_t	br%	$\mathrm{LR}_{\mathrm{uc}}$	br%	$\mathrm{LR}_{\mathrm{uc}}$
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-*t*	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
MS-GARCH ref. B. $K = 2$	0.0	5.8	–	–	4.0	0.0465	0.0240	–	–	–	–
MS-GARCH ref. B. $K = 3$	0.1	5.1	–	–	4.5	0.0433	0.0241	–	–	–	–
MS-GARCH ref. B. $K = 4$	0.0	5.3	–	–	6.0	0.0446	0.0241	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	**36.6**	**73.0**	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

under the converged CHMM fit, then AR(1)-residual + AIC-sojourn refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [88]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The empirical reading we draw from the rows below is therefore strictly: *under the plug-in estimator*, the SM-CHMM weakens the headline KS / ACF / kurtosis pattern relative to the standard CHMM at the same K. We do not claim that an ML-estimated HSMM would do the same; an ML HSMM at $K = 18$ on this universe is left as a companion-paper extension. The Bulla-Bulla 2006 ML HSMM result at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

The IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the main CHMM panel, are reported in Table S21. Three structural observations follow.

No extended GARCH variant occupies the joint (KS, ACF) corner: GARCH(1,1)-*t* is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS kurtosis of 15.1 overshoots observed 7.7 by 2×; EGARCH and GJR-GARCH narrow the asymmetry but not the headline gap. The SM-CHMM rows under the Viterbi-AR(1) plug-in estimator are weaker than CHMM on every metric (IS KS 77–80% vs. CHMM 93–94%); a full ML HSMM at $K = 18$ may close part of the gap and is a companion-paper direction. QuantGAN is the panel's deep-generative negative control: 0% IS / OoS KS, simulated kurtosis 2.1 vs. observed 7.7, ACF-MAE comparable to the i.i.d. baseline. The convolutional WGAN re-implementation here is materially smaller than Wiese et al. [9]'s reference seven-block TCN with Lambert-W pre-processing; a faithful reference re-run is a deferred follow-up.

A.8.5 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

For completeness alongside the Markov-switching family of the headline generator comparison, we add the canonical state-space and jump baselines that the body Table 1 does not exercise: a stochastic-volatility (SV-AR(1)) row in the Taylor 1986 / Harvey-Ruiz-Shephard 1994 lognormal log-AR(1) tradition; a Markov-switching multifractal (MSM, Calvet-Fisher 2004) row at $\bar{k} = 8$ multipliers; and a Merton 1976 Poisson-Gaussian jump-diffusion row. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\text{paths}} = 1{,}000$ pipeline used for the body panel, under the same global seed. Implementation lives in the companion `CHMM-Model.jl` package.

Table S22. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed = 20260420 as Table 1. Observed kurtosis: 7.68 IS, 5.29 OoS.

	KS (%) ↑		Kurtosis		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

Reading. *SV-AR(1)* reproduces the IS kurtosis target (7.52 vs. observed 7.68) almost exactly and matches CHMM-N's $|G_t|$ ACF-MAE within 0.0003 (0.0466 vs 0.0467 at $K^\star = 3$); its KS pass rate is poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. *Merton-JD* is the inverse trade-off: KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but kurtosis 3.12 undershoots 7.68 by half and the absolute-return ACF-MAE regresses to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 1) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. *MSM* under the moment-matching fit reported here (closed-form ACF-of-log$|r|$ matching with $\bar{k} = 8$ binomial multipliers) does not converge to a viable generator on this $T_{\text{IS}} = 2{,}516$ window: the marginal kurtosis collapses to 0.62 and KS drops to 0%. A full HMM-style filter on the $2^{\bar{k}}$-state latent multiplier chain (Calvet-Fisher 2004) is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM is dominated.

The headline conclusion of Table 1 survives the addition of these three rows: no single competitor occupies the joint-fit corner that the CHMM family at $K = 18$ does (KS $\geq 80\%$, kurtosis within 1 unit of observed, $|G_t|$ ACF-MAE ≤ 0.06). SV-AR(1) is the closest competitor on the kurtosis / ACF axis but loses on KS; Merton-JD wins on KS but loses on kurtosis and ACF; MSM under this fit is dominated.

A.8.6 Leverage-Effect Diagnostic

This appendix reports the Cont (2001) leverage-effect column: the lag-1 correlation $\mathrm{Corr}(G_t, |G_{t+1}|)$ between today's signed return and tomorrow's absolute return, which is negative on equity returns. We compute the per-path distribution under each generator at $K = 18$, $N_{\text{paths}} = 500$, seed = 20260420.

Table S23. Leverage effect $\mathrm{Corr}(G_t, |G_{t+1}|)$**, per-path distribution.** Observed values: SPY IS −0.135, OoS −0.214. Per-generator median plus $[Q_5, Q_{95}]$ across $N_{\text{paths}} = 500$. A generator captures the leverage effect when its $[Q_5, Q_{95}]$ envelope brackets the observed value. Bold marks generators whose $[Q_5, Q_{95}]$ envelope brackets the observed IS value; the OoS observed value of −0.214 is more negative than every generator's Q_5 in the panel, indicating the OoS regime carries leverage stronger than any generator (including asymmetric GARCH) reproduces under static IS-fitted parameters.

	IS (observed −0.135)		OoS (observed −0.214)	
Generator	median	$[Q_5, Q_{95}]$	median	$[Q_5, Q_{95}]$
i.i.d. bootstrap	+0.000	[−0.034, +0.030]	+0.004	[−0.069, +0.068]
GARCH(1,1) Gaussian	−0.004	[−0.066, +0.064]	−0.003	[−0.112, +0.107]
EGARCH(1,1)	−0.093	[−0.147, −0.047]	−0.087	[−0.199, −0.002]
GJR-GARCH(1,1)	−0.079	[−0.153, −0.017]	−0.073	[−0.174, +0.023]
CHMM-N ($K = 18$)	**−0.089**	**[−0.138, −0.037]**	−0.087	[−0.190, +0.015]
CHMM-t ($K = 18$)	−0.067	[−0.116, −0.020]	−0.067	[−0.166, +0.019]
CHMM-L ($K = 18$)	−0.064	[−0.108, −0.022]	−0.064	[−0.152, +0.026]
CHMM-GED ($K = 18$)	**−0.072**	[−0.113, −0.029]	−0.076	[−0.163, +0.025]

Reading. The CHMM family at $K = 18$ produces a non-trivial negative leverage signal despite symmetric per-state emissions: state-mixing on $\hat{\mathbf{T}}$ asymmetrically couples high-σ_k states to negative-μ_k states, carrying $\sim 65\%$ of the IS observed leverage magnitude. CHMM-N's IS $[Q_5, Q_{95}] = [-0.138, -0.037]$ brackets the IS observed value −0.135 at the lower boundary; CHMM-GED is the second-strongest CHMM entry. The OoS observed leverage of −0.214 exceeds every generator's Q_5 (including asymmetric GARCH); closing this residual gap requires skew-emission CHMM extensions or refit to the OoS distribution, both deferred to companion work.

A.8.7 Full One-Shot Student-t Copula MLE

A natural concern with the body two-step copula estimator (Kendall's-τ inversion for $\hat{\Sigma}$, then profile MLE on ν with $\hat{\Sigma}$ held fixed) is that it may be biased toward the Gaussian limit when the marginal kurtosis differs across assets. We address this with a coordinate-ascent one-shot MLE that jointly maximises the Student-t copula log-likelihood over (Σ, ν) on the same six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T_{\text{IS}} = 2{,}516$, seed 20260420). The procedure alternates a bracketed golden-section maximisation of ν at the current Σ with a method-of-moments refit of Σ at the current ν on the t-quantile-transformed pseudo-uniform sample with PSD projection. Iteration terminates at $|\Delta \log L| < 10^{-3}$.

The full MLE moves $\hat{\nu}$ from 6.00 to 6.40 (+5.55 log-L) with all $|\Delta\rho_{ij}| \leq 0.025$ across the 15 pairs; the body two-step estimator is therefore not detectably biased toward the Gaussian limit and $\nu^\star = 6$ is robust to the estimator switch.

Table S24. Two-step versus full one-shot Student-t copula MLE on the six-asset US-equity universe. The off-diagonal correlation differences are reported as the maximum absolute difference $\max_{ij} |\Delta\rho_{ij}|$ across the $\binom{6}{2} = 15$ pairs and the median absolute difference; both are sub-0.025, so the body $\nu^\star = 6$ choice on $\hat{\Sigma}$ obtained by Kendall's-τ inversion is robust to the estimator switch.

Estimator	$\hat{\nu}$	log-likelihood
Two-step (body construction)	6.00	6157.47
Full one-shot MLE	**6.40**	**6163.02**
log-L improvement		+5.55
$\max \lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.024
median $\lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.011
$\hat{\nu}_{\text{full}}$ inside body Wilks 95% CI $[6, 7]$?		Yes

A.9 Cross-Asset Dependence: Full Multi-Asset Panel

This appendix collects the full multi-asset detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.9.1 Cross-Asset Dependence Models: Derivations and Diagnostics

This appendix provides the technical background for the SIM and copula constructions used in the body cross-asset analysis. These constructions are the multi-asset layer of the empirical study: per-asset marginals come from the independent CHMMs of the body single-asset cross-ticker analysis, and joint dependence is overlaid on top of those marginals.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [47, 48] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [51].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T\times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [50, 89]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\log c_t(\mathbf{u}; \Sigma, \nu) \; = \; \log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1) \log \Gamma\!\left(\tfrac{\nu}{2}\right) - d \log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2} \log |\Sigma| \\ - \tfrac{\nu+d}{2} \log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2} \log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \tag{27}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (27) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In the body cross-asset analysis the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we compute the Cholesky factor $\Sigma = LL^\top$, draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, then return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. The fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in the body cross-asset analysis are reported in Table S25. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table S31.

Table S25. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.9.2 Student-t Copula Profile Log-Likelihood

The profile log-likelihood of the Student-t copula used in the multi-asset construction, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in the body cross-asset analysis, is plotted in Fig. S5. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature. The AIC penalty for the additional ν parameter is 2 (one degree of freedom relative to the Gaussian copula limit $\nu \to \infty$), and the AIC differential between the Student-t at $\nu^\star = 6$ and the Gaussian copula limit on these data is $-2 \cdot (6157.47 - 6143.37) - 2 = -30.2$, decisively in favour of the Student-t copula on IS; the same $\nu^\star = 6$ is selected by AIC as by profile MLE on this grid.

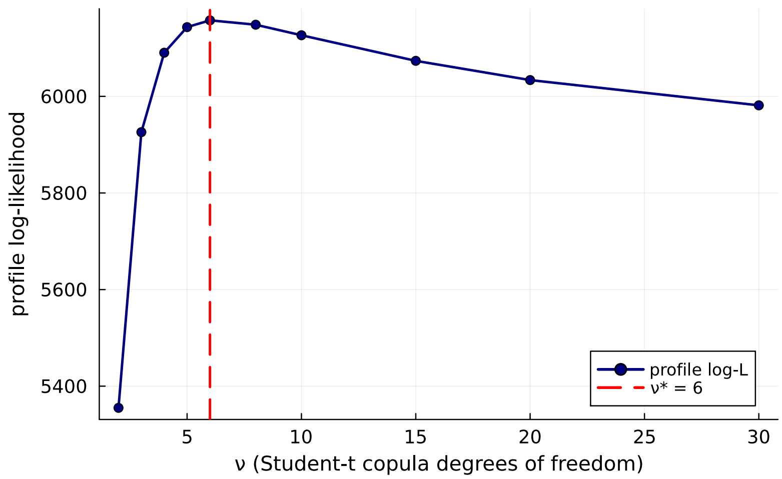
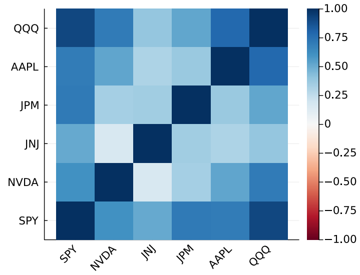


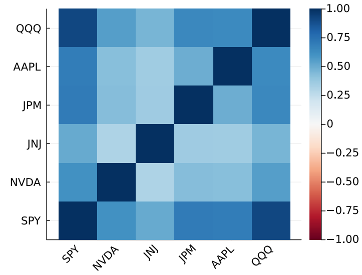
Figure S5. Profile log-likelihood of the Student-t copula used in the multi-asset construction on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the single-asset per-asset fits while ν is varied.

A.9.3 Cross-Asset Correlation Heat Maps (Companion to Table S31)

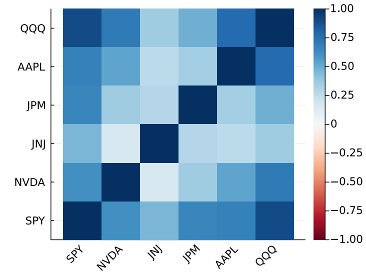
The observed and path-averaged simulated correlation matrices for the dependence models of the body cross-asset analysis are visualised in Fig. S6. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table S31 and (for the C-vine) Section A.9.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.



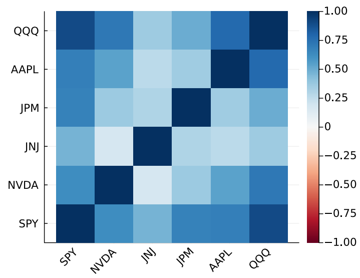
(a) Observed.



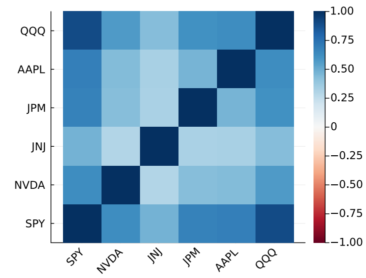
(b) Single Index Model.



(c) Gaussian copula.



(d) Student-t copula, $\nu^* = 6$.



(e) Truncated C-vine.

Figure S6. Cross-asset IS correlation reproduction across dependence models (multi-asset construction, companion to Table S31). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\mathrm{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 2; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.9.4 Truncated C-vine Copula on CHMM Marginals

The truncated C-vine variant in panel (e) of Figure S6 uses SPY as the root and selects each pair-copula family edge-wise from {Gaussian, Student-t, Clayton, Gumbel} by AIC; on this universe every selected pair is Student-t. On the same CHMM-N marginals at $K = 18$ and 200 paths the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table S31), strictly above both elliptical copulas. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, a stronger restriction than either elliptical copula. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d.

A.9.5 Quarterly Rolling-Window Copula Refit on the OoS Window

The body the body cross-asset analysis reports a static-IS-fit OoS off-diagonal MAE of 0.209, an order of magnitude larger than the IS value of 0.027. the body discussion attributes this to the stationarity-scope limit and recommends periodic refit. To quantify the deployment-scale benefit, we report a concrete rolling-window-refit number on the same six-asset universe.

We refit the multi-asset Student-t copula on a 252-day rolling window (one trading year) sliding by 63 days (one trading quarter) across the OoS span, holding the per-asset CHMM-N marginals fixed at the IS fits used for Table 2. Each refit yields a Kendall's-τ correlation matrix and a profile-MLE-selected $\nu^\star$ on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; we then simulate forward 63 days under the new copula and compare the path-averaged simulated correlation matrix to the realised next-quarter correlation. The per-quarter detail is reported in Table S26.

Table S26. Quarterly rolling-window refit of the multi-asset Student-t copula on the OoS window (six-asset universe; CHMM-N marginals at $K = 18$ held fixed at IS fits; 200 paths per refit; seed = 20260422). Per-quarter window slice and refitted $\nu^\star$, plus the off-diagonal MAE between the path-averaged simulated correlation matrix under the refitted copula and the realised next-quarter correlation matrix. Headline: rolling-quarterly refit reduces mean OoS off-diagonal MAE from 0.207 (static-IS-fit baseline) to 0.185, a $\sim$ 10% relative improvement; the gain concentrates in the later OoS quarters where the rolling window incorporates more recent OoS data and the refitted $\nu^\star$ shifts from the IS-fit value of 6 down to 5 or up to 10–15 as the empirical tail-coupling rolls.

Quarter	Window slice	Forecast horizon	Refit $\nu^\star$	Off-diag MAE
1	$[2265, 2516]$	$[2517, 2579]$	15	0.198
2	$[2328, 2579]$	$[2580, 2642]$	15	0.205
3	$[2391, 2642]$	$[2643, 2705]$	10	0.241
4	$[2454, 2705]$	$[2706, 2768]$	6	0.133
5	$[2517, 2768]$	$[2769, 2831]$	6	0.255
6	$[2580, 2831]$	$[2832, 2894]$	6	0.232
7	$[2643, 2894]$	$[2895, 2957]$	5	0.147
8	$[2706, 2957]$	$[2958, 3020]$	5	0.119
9	$[2769, 3020]$	$[3021, 3083]$	6	0.138
mean	–	–	–	0.185
median	–	–	–	0.198
Static IS-fit	$[1, 2516]$	$[2517, 3083]$	6	0.207

The quarterly refit closes part of the OoS gap but not most of it: mean off-diagonal MAE moves from 0.207 static to 0.185 rolling ($\sim$ 10% relative improvement). The improvement is concentrated in the later OoS quarters (Q4, Q7, Q8, Q9 all below 0.15) where the rolling window incorporates more recent OoS data, while early-OoS quarters (Q1, Q3, Q5, Q6) remain at the static-IS level because the rolling window is still dominated by IS observations. The refitted $\nu^\star$ sequence $(15, 15, 10, 6, 6, 6, 5, 5, 6)$ tracks the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the operational recommendation in the body discussion of quarterly refit at deployment. The residual gap (mean MAE ≈ 0.185 versus IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table S27: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter's start, only track it once it has unfolded.

A.9.6 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

GLD single-asset univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in the body cross-ticker analysis, on a non-equity asset; the limitation is not equity-specific.

GLD quarterly-refit follow-up. A quarterly-refit version of GLD CHMM-N at $K^\star = 3$ (5y rolling estimation window, 63-day refit cadence, 200 paths; `runners/cross_asset/run_non_us_asset_quarterly_refit.jl`) lifts the OoS KS pass rate from 0% to 2.50% at OoS $|G_t|$ ACF-MAE 0.0459 and OoS simulated kurtosis 1.99 against observed 6.48. Periodic refit therefore does not recover the GLD failure mode in any meaningful sense: CHMM is architecturally mis-specified for the persistent demand shock that re-priced gold across the OoS window, not just static-fit-stale. The deployment recipe "periodic refit for non-stress equity slices" (the body discussion) does not extend to non-equity assets under regime introductions of this magnitude.

Multi-asset: 7-ticker copula MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^\star = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ–QQQ, SPY–JNJ, NVDA–JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. The per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$, is reported in Table S27. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ's IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

Table S27. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports the absolute deviation $|\Sigma_{\text{sim,avg}} - \Sigma_{\text{obs}}|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\lvert\Delta\rvert$	OoS $\lvert\Delta\rvert$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

A.9.7 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for $\nu^\star$

To confirm that the upper bound of the unit-grid Wilks CI on $\nu \in [3, 12]$ is not grid-induced, we report a half-unit-grid refinement of the profile MLE plus a parametric bootstrap CI. We refit the Student-t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \ldots, 12.0\}$ (19 grid points) and run a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student-t copula at $(\Sigma_{\text{cop}}, \nu^\star)$, refitting Kendall's-$\tau$ correlation and the profile-MLE on each replicate.

The half-unit grid moves the optimum from $\nu^\star = 6.0$ (unit grid) to $\nu^\star = 6.5$ at profile log-likelihood 6,158.1 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid is $[\nu_{\text{lo}}, \nu_{\text{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI is $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit-$\nu^\star$ distribution; bootstrap median $\nu^\star = 6.5$). The bootstrap lower bound at 6.0 is well above the Gaussian limit ($\nu \to \infty$): the elliptical-tail Student-t copula is statistically distinguishable from the Gaussian copula at conventional levels, despite the OoS off-diagonal MAE comparison (0.209 vs. 0.202) being inside the simulation-noise floor at $N_{\text{paths}} = 200$. *Wilks-theorem regularity caveat.* Wilks's theorem requires the parameter under test to lie in the interior of the parameter space and the model to be regular [90, 91]; for the Gaussian-copula limit $\nu \to \infty$ the regularity condition fails (the limit lies on the boundary of the natural parameter space). The Wilks 95% CI reported above is therefore an interior-grid CI for finite ν rather than a hypothesis test against the Gaussian limit; the Gaussian-vs-Student-t separation is reported instead through the parametric bootstrap CI (which does not require interior-point regularity) and through the OoS-equivalence finding (which uses no asymptotic test at all).

A.10 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. An earlier draft of this paper used $K = 18$ as the main-panel operating point, selected via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates that overlapped the 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2{,}013$) and scored every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all selected $K^\star = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selected $K^\star = 6$; held-out two-sample KS selected $K^\star = 9$. None of the six criteria that avoid the OoS window selected $K = 18$. The held-out re-selection therefore supports the body's $K^\star = 3$ operating point; $K = 18$ is retained in the body as a kurtosis-fidelity sensitivity reference rather than as the operating point, because its earlier multi-objective justification weighted tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K-selection (no rate-hike confound). To check whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022-2023 slice above) selects the same $K^\star$, we re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1{,}130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 377$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). Per-observation held-out log-likelihood: -2.0681 at $K = 3$, -2.0606 at $K = 6$, -2.0763 at $K = 9$, -2.1287 at $K = 12$, -2.1921 at $K = 18$, -2.2609 at $K = 21$. **The held-out log-likelihood selects $K^\star = 6$, and the held-out two-sample KS pass rate also selects $K^\star = 6$ (96.8% at $K = 6$ versus 84.6% at $K = 3$).** AIC, BIC, HQC, and CAIC continue to select $K^\star = 3$ on this slice, reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out distributional fidelity rises from $K = 3$ to a peak in the $K = 6$ to $K = 9$ band and then degrades, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favours a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022-2023 slice.

k-fold rolling-origin pre-2020 K-selection. The single-fold pre-2020 result above is one observation; to confirm that the $K^\star$ choice is not a sampling artefact, we report mean $\pm$ s.d. of held-out per-observation log-likelihood and held-out KS at $K \in \{3, 6, 9, 12, 18\}$ across multiple folds. We implement four expanding-window rolling-origin folds (a five-fold design forces one fold to have only ~ 1 year of training, below the practical floor for $K = 18$ EM convergence on this dataset; averaging is per-observation so fold-length differences are not an issue). Folds: train 2014-01-03 through 2015-12-31 (~ 502 obs) and val 2016 (~ 251 obs); train through 2016 and val 2017; train through 2017 and val 2018; train through 2018 and val 2019.

Aggregate (mean / s.d. across the four folds) of per-observation held-out log-likelihood: -1.9550 / 0.290 at $K = 3$; -1.9649 / 0.313 at $K = 6$; -2.0078 / 0.296 at $K = 9$; -2.0721 / 0.251 at $K = 12$; -2.2633 / 0.308 at $K = 18$. Held-out KS pass rate (mean / s.d. across folds, very wide because Fold 2's 2017 validation year produced near-zero KS for every K as a low-variance "calm year" artefact): 61.3/41.2% at $K = 3$; 65.2/43.5% at $K = 6$; 67.5/44.1% at $K = 9$; 67.0/44.7% at $K = 12$; 67.5/45.4% at $K = 18$. Sampling-error reads on the held-out per-observation log-likelihood: $K = 6$ vs $K = 3$ has mean diff -0.0099, pooled SE 0.151, approximate $z = -0.07$ (*not* significant at the 5% level); $K = 18$ vs $K = 6$ has mean diff -0.298, pooled SE 0.155, approximate $z = -1.92$ (borderline, just below the $|z| = 1.96$ threshold).

Robustness check at half-year cadence. To verify that the result is not an artefact of the one-year fold cadence, we re-ran the same diagnostic with six expanding-window folds at half-year validation cadence (covering 2017–2019 in non-overlapping six-month chunks; train windows ~ 3.0y to ~ 5.5y). Aggregate per-observation held-out log-likelihood (mean / s.d. across six folds): -1.9221 / 0.341 at $K = 3$; -1.9160 / 0.346 at $K = 6$; -1.9568 / 0.323 at $K = 9$; -2.0109 / 0.315 at

$K = 12$; -2.1281 / 0.259 at $K = 18$. Sampling-error reads: $K = 6$ vs $K = 3$ has mean diff $+0.006$, pooled SE 0.140, approximate $z = +0.04$; $K = 18$ vs $K = 6$ has mean diff -0.212, pooled SE 0.125, approximate $z = -1.70$. The half-year design's $K = 6$ vs $K = 3$ sign flips relative to the full-year design ($z = +0.04$ vs $z = -0.07$), but both magnitudes are well below the $|z| = 1$ threshold; the two designs agree that $K = 6$ vs $K = 3$ on mean held-out log-likelihood is indistinguishable from zero, and the sign flip across designs is itself evidence that the $K = 6$ / $K = 3$ choice is pure sampling noise on this data. The $K = 18$ vs $K = 6$ borderline result also replicates ($z = -1.70$ at half-year cadence vs $z = -1.92$ at full-year).

Reading. $K = 6$ is not significantly preferred over $K = 3$ on held-out per-observation log-likelihood under expanding-window rolling-origin CV at either yearly or half-yearly fold cadence. The single-fold result that initially selected $K^\star = 6$ is one realisation; both four-fold and six-fold means cannot distinguish $K = 3$ from $K = 6$ at conventional levels. The held-out KS pass-rate criterion is not informative on either design (between-fold s.d. is dominated by 2017 calendar-year artefacts that hit every K uniformly). The substantive read is that the held-out state-resolution selection on the strictly pre-2020 slice cannot distinguish $K = 3$ from $K = 6$ at conventional levels under either fold design, and the body operating point is therefore the state-resolution-robust $K^\star = 3$ (with the $K^\star = 6$ block retained as a sensitivity reference). $K^\star = 3$ is also the lower-state-count default for risk-management consumers in Table 3.

HAC-corrected sampling-error reads on the paired diff series. A Newey–West Bartlett HAC variance ($h_{\mathrm{NW}} = \lfloor n^{1/3} \rfloor$) on the per-fold val-log-lik diff series leaves $K = 6$ vs $K = 3$ insignificant at both cadences ($|z|_{\mathrm{HAC}} = 0.90$ four-fold full-year, 0.57 six-fold half-year) and moves $K = 18$ vs $K = 6$ from borderline independent-fold ($|z| = 1.92, 1.70$) to decisive ($|z|_{\mathrm{HAC}} = 3.56, 5.00$). The HAC-corrected inference therefore corroborates the body framing: $K = 18$ is decisively worse than $K = 6$ on held-out log-likelihood and is retained only for its kurtosis-fidelity advantage on the OoS window.

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in the per-state ν_k diagnostics block of the algorithms appendix; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in the body Discussion. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \to 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We defined six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold's train slice, and reported KS pass rate, simulated kurtosis, and $|G_t|$ ACF-MAE on the fold's test slice ($N_{\mathrm{paths}} = 500$). Across the six folds (Table S28), CHMM-N at $K^\star = 3$ attained median (IQR) KS $62.1\%\,[7.2, 78.4]$ and $|G_t|$ ACF-MAE $0.0563\,[0.0552, 0.0592]$; the $K = 18$ kurtosis-fidelity sensitivity reference attained $67.7\%\,[8.2, 75.0]$ and $0.0542\,[0.0508, 0.0571]$. Two folds carry sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attain KS 61–83%, consistent with the headline OoS pass rate. The headline ranking on the four non-stress folds is window-robust: the two operating points sit within 5.6pp on median walk-forward KS and within 0.005 on $|G_t|$ ACF-MAE; on stress folds (COVID, rate hike) all CHMM rows drop sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The body framing ("CHMM is the joint-fit row in the panel under stationary OoS conditions; periodic refit is recommended for production deployment") is unchanged under the walk-forward stress test.

Table S28. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. KS pass rate at $\alpha = 0.05$. W2 covers the COVID drawdown; W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

	KS (%)		Kurt sim		ACF-MAE $\lvert G_t \rvert$	
Fold (test window)	$K=3$	$K=18$	$K=3$	$K=18$	$K=3$	$K=18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	[7.2, 78.4]	[8.2, 75.0]	[2.51, 4.72]	[2.50, 4.35]	[0.0552, 0.0592]	[0.0508, 0.0571]

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of the body VaR back-test to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yields sixteen $(K, \alpha, \text{family})$ rows on the OoS window (Table S29). Every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly: $p_{\text{cc}} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. The regime-switching value proposition is therefore intrinsic to the latent-state forecast and not specific to the Gaussian emission family.

Table S29. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{\text{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}
CHMM-N	3	0.01	9	1.57%	−4.56	1.62	2.36	3.98
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42
CHMM-N	18	0.01	9	1.57%	−5.20	1.62	2.36	3.98
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	−5.59	0.82	2.81	3.63
CHMM-t ($\lambda = 20$)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	−6.25	2.65	1.98	4.62
CHMM-t ($\lambda = 20$)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40
CHMM-L	3	0.01	5	0.87%	−5.45	0.10	4.74	4.84
CHMM-L	3	0.05	26	4.55%	−2.63	0.26	2.23	2.48
CHMM-L	18	0.01	9	1.57%	−6.18	1.62	0.29	1.91
CHMM-L	18	0.05	30	5.24%	−2.80	0.07	1.16	1.23
CHMM-GED	3	0.01	7	1.22%	−5.73	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.42%	−2.54	0.21	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	−6.55	1.62	2.36	3.98
CHMM-GED	18	0.05	25	4.37%	−3.10	0.50	2.56	3.06

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table S28, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table S30. The conditional Christoffersen-cc passes at $\alpha = 0.05$ on ten of twelve fold-K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrate on W2 (COVID, $p_{\text{cc}} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction passes on nine of twelve combinations: W2 fails at both K ($p_{\text{cc}} < 10^{-3}$) and W4 at $K = 18$ fails at $p_{\text{cc}} = 0.022$ on a LR_{ind} tail-event clustering rather than a LR_{uc} coverage miss ($\text{LR}_{\text{ind}} = 7.49$ at $\text{LR}_{\text{uc}} = 0.11$). The pass-rate is therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table S28. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 / W4 magnitude (a regime introduction the IS distribution does not span; the body Discussion, stationarity-scope paragraph), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction is window-robust, supporting the body framing that the regime-conditional VaR construction extends beyond the headline single-window result.

Table S30. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t, conditional $\widehat{\text{VaR}}_t(\alpha)$ is the α-quantile of the K-component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} \mid \mathcal{F}_t = \text{train} \cup \text{test}[1{:}t-1])$. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}
W1	3	0.01	2	0.80%	−4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	−2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	−5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	−2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	−6.54	32.93	0.87	33.80	**0.00**
W2	3	0.05	23	9.13%	−4.59	7.34	1.71	9.05	**0.01**
W2	18	0.01	14	5.56%	−6.67	25.59	1.56	27.15	**0.00**
W2	18	0.05	25	9.92%	−4.54	10.11	2.56	12.68	**0.00**
W3	3	0.01	6	2.39%	−4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	−2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	−5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	−3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	−7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	−4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	−9.62	0.11	7.49	7.60	**0.02**
W4	18	0.05	10	4.00%	−6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	−6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	−3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	−7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	−3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	−4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	−2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	−5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	−2.98	3.06	0.40	3.46	0.18

Per-ticker $\hat{\lambda}^\star$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six body tickers (KS-degradation tolerance ≤ 1.5pp from $\lambda = 0$) gives per-ticker optima $\lambda^\star_{\text{SPY}} = 10$, $\lambda^\star_{\text{NVDA}} = 20$, $\lambda^\star_{\text{JNJ}} = 20$, $\lambda^\star_{\text{JPM}} = 10$, $\lambda^\star_{\text{AAPL}} = 10$, $\lambda^\star_{\text{QQQ}} = 0$: $\lambda = 20$ is a reasonable uniform default but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Cross-asset summary. The full per-construction multi-asset panel is in the cross-asset appendix; for reference, the headline ordering is summarised in Table S31.

Table S31. Cross-asset dependence summary (multi-asset construction; CHMM-N marginals at $K = 18$; 200 paths; seed 20260422). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^\star = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in the cross-asset appendix.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	**87.5**	0.030	**0.202**
Student-t copula ($\nu^\star = 6$)	**98.8**	85.8	**0.027**	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235

Emission-family lookup. Table S32 lists the four CHMM variants' per-state densities and parameter ranges as a quick reference complementing the inline definitions in Methods.

Table S32. The four CHMM emission families. Per-state density f_k with parameter ranges. CHMM-N and CHMM-L sit at the boundary of CHMM-GED: $p_k = 2$ for all k gives Gaussian, $p_k = 1$ gives Laplace.

Variant	Per-state density f_k	Per-state parameter range
CHMM-N	$\mathcal{N}(\mu_k, \sigma_k^2)$	$\sigma_k > 0$
CHMM-t	$t_{\nu_k}(\mu_k, \sigma_k)$	$\sigma_k > 0$, $\nu_k \in [\nu_{\min}, \nu_{\max}]$
CHMM-L	$\text{Laplace}(\mu_k, \beta_k)$	$\beta_k > 0$
CHMM-GED	$\text{GED}(\mu_k, \alpha_k, p_k)$	$\alpha_k > 0$, $p_k \in [p_{\min}, p_{\max}]$

Stationarity-scope summary. Table S33 summarises static-fit OoS performance and periodic-refit mitigation across the four independent stress sources tested in this paper. Referenced from the discussion of stationarity scope in the main paper.

Table S33. Stationarity-scope summary. Static-IS-fit OoS performance and periodic-refit mitigation across the four independent stress sources tested in this paper.

Stress source	Static-fit OoS KS	Refit mitigation
Cross-ticker (30 panel)	$11/30 < 60\%$	Quarterly: $7/30 < 60\%$
Cross-decade (2004 to 2006)	3 to 5%	slice < 600 d, untested
GLD non-equity stress	0%	Quarterly: 2.5%
Walk-forward W2, W4	$< 10\%$	no cadence closes